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Supplementary Material for Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT This supplement records the detailed literature matrix, electrical conventions, mathematical proofs, Fisher-information analysis, Bayesian estimator construction, frozen experimental contracts, and secondary results supporting the main article. It distinguishes the independently evaluated state component and implemented batch event estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. The latter remains unexecuted. This supplement does not add empirical claims or change any test denominator. Every empirical table and figure is generated from the same hashed evidence snapshot used by the main manuscript; separate deterministic tables audit stated algebraic identities.

INDEX TERMS Bayesian inference, Fisher information, functional observability, phasor measurement units, reproducibility, state estimation, event-source inference.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. "Held source" means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_\theta \dot{\theta}). \tag{1}$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential states remain continuous unless a simulator-specific reset is declared.

With $Y_{\mathrm{bus}} = G + \mathrm{j}B$ and $V_i = |V_i|e^{\mathrm{j}\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j|(G_{ij}\cos\vartheta_{ij} + B_{ij}\sin\vartheta_{ij}), \tag{2}$$

$$Q_i = |V_i| \sum_j |V_j|(G_{ij}\sin\vartheta_{ij} - B_{ij}\cos\vartheta_{ij}). \tag{3}$$

For branch $e = (i,j)$ with complex tap t_e, series admittance y_e, and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \tag{4}$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

Positive-sequence phasors define the balanced model. Negative- and zero-sequence components remain integrity

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic *et al.* [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho *et al.* [2]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu *et al.* [3]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo *et al.* [4]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue *et al.* [5]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshtagh *et al.* [6]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey *et al.* [7]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [8]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li *et al.* [9]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow *et al.* [10]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [11]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus *et al.* [12]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai *et al.* [13]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
Main article	Hidden voltage field	All load supports with $K \leq 2$; historical eight-label ontology	DAE hypotheses and quadratic physical likelihood	Eight PMU buses	Physical bank has no source-label head	Physical/integrity variables separate

or unbalance evidence. Ideal continuous-frequency channels satisfy

$$f_i^{\mathrm{phys}} = f_0 + \frac{1}{2\pi}\frac{d\angle V_i}{dt}, \qquad \mathrm{ROCOF}_i^{\mathrm{phys}} = \frac{1}{2\pi}\frac{d^2\angle V_i}{dt^2}, \tag{5}$$

but an operational likelihood must reproduce the causal filtering of the PMU backend.

The preceding equations define the ideal physical quantities f_i^{phys} and $\mathrm{ROCOF}_i^{\mathrm{phys}}$ implied by the network trajectory. Reported quantities f_i^{meas} and $\mathrm{ROCOF}_i^{\mathrm{meas}}$, however, are outputs of a dynamic frequency-estimation process and can exhibit disturbance- and estimator-dependent transient error and latency. A prior benchmark by the authors compared 16 dynamic frequency estimators spanning six practical groups over 34 disturbance scenarios and quantified these trade-offs under ramps, phase jumps, harmonics, ringdown, and composite IBR disturbances [14]. We therefore distinguish

$$\begin{bmatrix} f_i^{\mathrm{meas}} \\ \mathrm{ROCOF}_i^{\mathrm{meas}} \end{bmatrix} = \begin{bmatrix} f_i^{\mathrm{phys}} \\ \mathrm{ROCOF}_i^{\mathrm{phys}} \end{bmatrix} + \boldsymbol{\varepsilon}_{fr,i}, \qquad \mathrm{Cov}(\boldsymbol{\varepsilon}_{fr,i}) = \boldsymbol{\Omega}_{fr}. \tag{6}$$

Because phasor, frequency, and ROCOF outputs may be estimated from overlapping underlying waveform or phasor information, their errors need not be conditionally independent. An augmented likelihood therefore permits cross-covariance

$$\boldsymbol{\Omega}_{\mathrm{aug}} = \begin{bmatrix} \boldsymbol{\Omega}_{VI} & \boldsymbol{\Omega}_{VI,fr} \\ \boldsymbol{\Omega}_{fr,VI} & \boldsymbol{\Omega}_{fr} \end{bmatrix}; \tag{7}$$

neither the magnitude of $\boldsymbol{\Omega}_{VI,fr}$ nor a block-diagonal approximation is assumed without calibration. Frequency and ROCOF are retained as candidate channels for an augmented sparse-PMU likelihood. Their incremental source-discrimination value, covariance structure, and effect on localization delay are not assumed here and require the dedicated VI, $VI + f$, $VI + \mathrm{ROCOF}$, and $VI + f + \mathrm{ROCOF}$ ablation.

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r, hidden-voltage relative error is

$$\mathrm{HVRE}^{(r)} = \frac{1}{T_r|\mathcal{U}|}\sum_{k=1}^{T_r}\sum_{i\in\mathcal{U}}\frac{|\widehat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \tag{8}$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the in-

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	ΔP_G, ΔQ_G, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	ΔP_L, ΔQ_L, ZIP convention	continuous input-driven response

dependent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

This appendix develops the mathematical chain used by the paper. It begins with the sampled hybrid event map, separates state-functional recovery from event-source separation, derives the relevant Fisher information after physical nuisance is removed, and then states the finite-horizon and temporal limits. The results are local unless a statement explicitly says otherwise.

A. REGULARITY AND STATISTICAL CONTRACT

Let $u = \mathrm{col}(x, z)$ collect differential and algebraic coordinates and consider the descriptor model

$$E\dot{u} = F(u, p), \qquad y = h(u, p). \tag{9}$$

The analysis uses four assumptions.

1) The reference angle has been fixed and the algebraic Jacobian is nonsingular on the local mode under study, so a regular index-one chart exists between discrete events.
2) The scheduled event time is fixed. In a neighborhood of zero amplitude, the complete callback–solver–sampling map is twice continuously differentiable.
3) Every covariance used as a metric is positive definite on the retained measurement subspace. Candidate-dependent covariances retain their log determinants in likelihood comparisons.
4) The support set is finite and all Bayesian priors are proper. Exact signed-amplitude equivalences use a neighborhood of zero within each support subspace; constrained physical amplitudes instead use their actual difference set.

These assumptions distinguish the local theory from general descriptor left invertibility, state-triggered saltation, and global nonlinear identifiability.

B. SIMULATOR-CONSISTENT EVENT SENSITIVITIES

For a fixed-time, impulse-free event, the differential state is continuous, but the algebraic state must satisfy the post-event constraint. This physical right limit must be distinguished from the values stored inside a simulator callback and from the first sampled frame. The following result makes that distinction explicit.

Theorem 1 (Local consistency-jump expansion): Let $x_0 = x(\tau^-)$ and suppose $g_+(x_0, z_0, 0) = \mathbf{0}$. If g_+ is three times continuously differentiable and g_z is nonsingular at $(x_0, z_0, 0)$, there is a unique local consistent branch

$$z_+(a) = z_0 + az_1 + \tfrac{1}{2}a^2 z_2 + O(a^3), \tag{10}$$

where all derivatives below are evaluated at the nominal right-limit mode and

$$z_1 = -g_z^{-1} g_a, \tag{11}$$

$$z_2 = -g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{12}$$

The corresponding differential state satisfies $x(\tau^+) = x(\tau^-)$, and its first event-amplitude sensitivity starts with

$$\dot{x}_a(\tau^+) = f_a - f_z g_z^{-1} g_a \equiv B_{r,a}. \tag{13}$$

Its second-order initial forcing is

$$\dot{x}_{aa}(\tau^+) \equiv B_{r,aa} = f_{zz}[z_1, z_1] + 2f_{za}z_1 + f_{aa} - f_z g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{14}$$

a: Proof.

The implicit-function theorem applied to $g_+(x_0, z_+(a), a) = \mathbf{0}$ gives a unique smooth local branch. Differentiating once gives $g_z z_1 + g_a = \mathbf{0}$, which yields (11). Differentiating a second time gives $g_z z_2 + g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa} = \mathbf{0}$, which yields (12). An impulse-free fixed-time switch does not reset the energy-storage coordinates, so the differential state is continuous. Differentiating $\dot{x} = f_+(x, z_+(a), a)$ at the right limit, with $x_a(\tau^+) = \mathbf{0}$, gives (13). A second differentiation, using $x_{aa}(\tau^+) = \mathbf{0}$ and (12), gives (14). □

The theorem also defines an implementation-independent consistency diagnostic. If a callback changes the parameter but temporarily retains z_0, its algebraic residual is

$$r_{\mathrm{cb}}(a) = g_+(x_0, z_0, a) = ag_a + \tfrac{1}{2}a^2 g_{aa} + O(a^3). \tag{15}$$

Substituting the zeroth-, first-, and second-order approximations to (10) produces constraint residuals of orders $O(a)$, $O(a^2)$, and $O(a^3)$, respectively. Central finite differences of an independently solved algebraic branch can therefore test both derivatives and the expected convergence orders without relying on undocumented callback storage.

For a differentiable PMU map $y = h(x, z, a)$, the direct right-limit signature is

$$y_a(\tau^+) = h_a - h_z g_z^{-1} g_a, \qquad \Delta y_{\mathcal{P}} = aM_{\mathcal{P}} y_a(\tau^+) + O(a^2). \tag{16}$$

This direct algebraic visibility is not the entire event signature. Even when (16) is nearly zero at the retained PMUs, (13) can generate a distinguishable dynamic response over later frames. Instantaneous invisibility therefore does not imply finite-horizon nonidentifiability.

The frozen artifacts begin at the first retained 30-Hz frame and do not record the callback microstate or an independently solved algebraic right limit. Accordingly, the paper treats the right-limit equations above as the physical contract, not as an observed fact about the current ANDES callback implementation. The relevant differentiable object for the existing response dictionary is the sampled map

$$\Psi_{T_s}^{\mathrm{sim}} : (u^-, p(a)) \mapsto u(\tau + T_s; a), \qquad T_s = 1/30\ \mathrm{s}. \tag{17}$$

For event coordinates a_i, define

$$L_i = \left.\frac{\partial \Psi_{T_s}^{\mathrm{sim}}}{\partial a_i}\right|_{a=\mathbf{0}}, \qquad L_{ij}^{(2)} = \left.\frac{\partial^2 \Psi_{T_s}^{\mathrm{sim}}}{\partial a_i \partial a_j}\right|_{a=\mathbf{0}}. \tag{18}$$

For $t \geq \tau + T_s$, the first and second sensitivities obey

$$E\dot{s}_i = F_u s_i + F_p p_i, \tag{19}$$
$$E\dot{s}_{ij} = F_u s_{ij} + r_{ij}, \tag{20}$$

where

$$r_{ij} = F_{uu}[s_i, s_j] + F_{up}[s_i, p_j] + F_{pu}[p_i, s_j] + F_{pp}[p_i, p_j] + F_p p_{ij}. \tag{21}$$

The initial conditions are $s_i(\tau + T_s) = L_i$ and $s_{ij}(\tau + T_s) = L_{ij}^{(2)}$. For the affine event parameterization, $p_{ij} = \mathbf{0}$.

Theorem 2 (Propagation of event-map mismatch): Suppose two second-order models have the same nominal trajectory, first-order sensitivities, parameter derivatives, and forcing in (20), but different second-order event-map initial conditions. Their difference $e_{ij} = s_{ij}^A - s_{ij}^B$ satisfies

$$E\dot{e}_{ij} = F_u e_{ij}, \qquad e_{ij}(t) = \Phi_E(t, t_1) e_{ij}(t_1) \tag{22}$$

on the consistent solution subspace, where $t_1 = \tau + T_s$.

b: Proof.

Subtract (20) for the two models. Every forcing term cancels by assumption, leaving the homogeneous descriptor variational equation. Applying its transition operator gives (22). □

When the retained PMU operator is locally affine, $y = H_y u + d$, the sampled response coefficients are

$$D_i = H_y s_i, \quad Q_i = \tfrac{1}{2} H_y s_{ii}, \quad Q_{ij} = H_y s_{ij}. \tag{23}$$

Thus all quadratic curvature comes from the physical solution and event map, not from PMU post-processing. This is why the dictionary must differentiate the same event semantics used to generate the trajectories.

C. EXACT FUNCTIONAL RECONSTRUCTION

Consider the finite-horizon linear experiment

$$Y = \mathcal{O}x + \varepsilon, \qquad \psi = Lx, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, R), \quad R \succ \mathbf{0}. \tag{24}$$

Theorem 3 (Functional reconstruction equivalence): The following statements are equivalent: (i) $\mathcal{O}x_1 = \mathcal{O}x_2$ implies $Lx_1 = Lx_2$; (ii) $\ker \mathcal{O} \subseteq \ker L$; and (iii) there is a linear map K such that $L = K\mathcal{O}$.

a: Proof.

If (i) holds and $v \in \ker \mathcal{O}$, the observations of v and $\mathbf{0}$ coincide, hence $Lv = \mathbf{0}$ and (ii) follows. If (ii) holds, define $K(\mathcal{O}x) = Lx$ on $\mathrm{range}\,\mathcal{O}$. The definition is independent of the representative because two representatives differ by a vector in $\ker \mathcal{O} \subseteq \ker L$. Extend K linearly to the observation space to obtain (iii). Finally, (iii) immediately implies (i). □

Let

$$G = \mathcal{O}^{\mathsf{T}} R^{-1} \mathcal{O}. \tag{25}$$

Theorem 4 (Best linear unbiased estimate of an estimable function): If the equivalent conditions of Theorem 3 hold, then

$$\widehat{\psi}_\star = L G^\dagger \mathcal{O}^{\mathsf{T}} R^{-1} Y, \tag{26}$$
$$\mathrm{Cov}(\widehat{\psi}_\star) = L G^\dagger L^{\mathsf{T}}. \tag{27}$$

It has minimum covariance, in the positive-semidefinite order, among all linear unbiased estimators of Lx [15].

b: Proof.

Because $v^{\mathsf{T}} G v = \|\mathcal{O}v\|_{R^{-1}}^2$, $\ker G = \ker \mathcal{O}$. Estimability gives $L = LG^\dagger G$, so taking expectation in (26) yields Lx. Direct covariance propagation gives (27). Let AY be any other linear unbiased estimator and set $T = A - LG^\dagger \mathcal{O}^{\mathsf{T}} R^{-1}$. Then $T\mathcal{O} = \mathbf{0}$ and the cross covariance vanishes because $LG^\dagger \mathcal{O}^{\mathsf{T}} T^{\mathsf{T}} = \mathbf{0}$. Consequently,

$$\mathrm{Cov}(AY) = \mathrm{Cov}(\widehat{\psi}_\star) + TRT^{\mathsf{T}} \succeq \mathrm{Cov}(\widehat{\psi}_\star). \tag{28}$$

If the kernel inclusion fails, two states induce the same data law but different targets, so no estimator can be unbiased for every state. □

The pseudoinverse has this interpretation only for estimable targets. In the nonlinear DAE, a pointwise kernel test is not sufficient: local factorization requires the condition to hold on a regular constant-rank neighborhood with connected local fibers [16].

D. FISHER INFORMATION WITH PHYSICAL NUISANCE

For a differentiable Gaussian model

$$r \mid \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{\Sigma}(\boldsymbol{\theta})), \tag{29}$$

the Fisher information contains both mean and covariance contributions. For scalar a,

$$I(a) = \boldsymbol{\mu}_a^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_a + \tfrac{1}{2} \mathrm{tr}(\boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}_a \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}_a). \tag{30}$$

The second term must be retained when event amplitude changes the covariance. The evaluated load bank freezes $\boldsymbol{\Sigma}$, so only the first term is used.

To expose nuisance geometry, consider

$$r = ad + N\boldsymbol{\nu} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{31}$$

where $\boldsymbol{\nu}$ may contain initial-state, operating-point, parameter, or integrity directions. Let $W^{\mathsf{T}}W = \boldsymbol{\Sigma}^{-1}$, $\overline{d} = Wd$, $\overline{N} = WN$, and

$$P_{\perp} = I - \overline{N}\overline{N}^{\dagger}. \tag{32}$$

Theorem 5 (Efficient Fisher information after nuisance profiling): For deterministic unknown $\boldsymbol{\nu}$, the efficient information for a is

$$I^{\mathrm{det}}_{a|\nu} = \overline{d}^{\mathsf{T}} P_{\perp} \overline{d} \tag{33}$$

$$= d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d - d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N(N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N)^{\dagger}N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d. \tag{34}$$

It is zero exactly when the event tangent lies in the observable nuisance span.

a: Proof.

After whitening, the score for a is $\overline{d}^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}}$. Nuisance scores span $\{v^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}} : v \in \mathrm{range}\,\overline{N}\}$. Their Fisher inner product is Euclidean because $\mathbb{E}[\overline{\boldsymbol{\varepsilon}}\overline{\boldsymbol{\varepsilon}}^{\mathsf{T}}] = I$. Removing the nuisance-score projection leaves $(P_{\perp}\overline{d})^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}}$, whose variance is (33). Expanding the projector by an SVD gives (34). □

If instead $\boldsymbol{\nu} \sim \mathcal{N}(m_{\nu}, \boldsymbol{\Gamma})$ with a proper prior, marginalization yields

$$r \mid a \sim \mathcal{N}(ad + Nm_{\nu}, \boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}}), \tag{35}$$

$$I^{\mathrm{Bayes}}_{a|\nu} = d^{\mathsf{T}}(\boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}})^{-1}d. \tag{36}$$

Woodbury's identity shows that (36) tends to the no-nuisance information as $\boldsymbol{\Gamma} \to \mathbf{0}$ and to (33) as a full-rank nuisance prior becomes diffuse. Thus zero deterministic efficient information is not a universal impossibility statement: a restrictive proper prior can still supply separation.

Corollary 1 (Pre-event information cannot reduce local event information): If pre-event data add nuisance precision $\boldsymbol{\Lambda}_{\mathrm{pre}} \succeq \mathbf{0}$, then

$$I_{a|\nu,\mathrm{pre}} = I_{aa} - I_{a\nu}(I_{\nu\nu} + \boldsymbol{\Lambda}_{\mathrm{pre}})^{-1}I_{\nu a} \tag{37}$$

is nondecreasing in $\boldsymbol{\Lambda}_{\mathrm{pre}}$.

b: Justification.

If $\boldsymbol{\Lambda}_2 \succeq \boldsymbol{\Lambda}_1 \succeq \mathbf{0}$, matrix inversion reverses the order. Premultiplication and postmultiplication by the cross-information blocks preserve the positive-semidefinite order, and subtraction reverses it once more. □

For a known source, known positive sign, and fixed covariance, the standardized matched statistic

$$T = \frac{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}r}{\sqrt{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d}} \tag{38}$$

is standard normal under H_0 and has mean $a\sqrt{I}$ under amplitude a. A level-α one-sided test has power $1-\Phi(z_{1-\alpha}-a\sqrt{I})$. Hence the amplitude required for power $1-\beta$ is

$$a_{\alpha,\beta} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{I}}. \tag{39}$$

This is a conditional known-source benchmark, not the threshold of the 137-support search. Its empirical use in the paper is limited to the exploratory source-wise detectability association.

E. FIRST-ORDER MULTI-EVENT SUPPORT GEOMETRY AND ONSET INFORMATION

Let two first-order source signatures be d_g and d_j under fixed precision $G \succeq \mathbf{0}$. Allow the competing source to refit its amplitude. The local profiled distance satisfies

$$b_{\star}(a) = a\frac{d_j^{\mathsf{T}}Gd_g}{d_j^{\mathsf{T}}Gd_j} + O(a^2), \tag{40}$$

$$\Delta^2_{g\to j}(a) = a^2 J_{g\to j} + O(a^3), \tag{41}$$

$$J_{g\to j} = d_g^{\mathsf{T}}Gd_g - \frac{(d_g^{\mathsf{T}}Gd_j)^2}{d_j^{\mathsf{T}}Gd_j} \geq 0. \tag{42}$$

Equation (42) is a pairwise local separation quantity, not the Fisher information for amplitudes conditional on the true support.

The pairwise formula extends to arbitrary supports. Let N_c be the unwhitened nuisance Jacobian, let $G_e = [g_1, \ldots, g_M]$ collect the event tangents, and define

$$A = WN_c, \quad F = WG_e, \quad R_N = I - AA^{\dagger}, \quad \overline{F} = R_N F. \tag{43}$$

Theorem 6 (Profile-likelihood diagnostic quotient): Consider the whitened local experiment

$$y = A\boldsymbol{\nu} + F\boldsymbol{\alpha} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \tag{44}$$

with deterministic unrestricted nuisance $\boldsymbol{\nu}$. Let $U_{\mathcal{D}}$ have orthonormal columns spanning $\mathcal{D} = \mathrm{range}(\overline{F})$, and define

$$z = U_{\mathcal{D}}^{\mathsf{T}}R_N y, \qquad d_{\mathcal{D}} = \dim\mathcal{D} = \mathrm{rank}(\overline{F}). \tag{45}$$

Then every profile-likelihood ratio over $\boldsymbol{\alpha}$ depends on y only through z:

$$\min_{\boldsymbol{\nu}} \|y - A\boldsymbol{\nu} - F\boldsymbol{\alpha}\|_2^2 = \|z - U_{\mathcal{D}}^{\mathsf{T}}\overline{F}\boldsymbol{\alpha}\|_2^2 + C(y), \tag{46}$$

where $C(y)$ is independent of $\boldsymbol{\alpha}$. The projector, quotient statistic up to an orthogonal change of coordinates, and $d_{\mathcal{D}}$ depend only on the nuisance span, not its parameterization.

a: Proof.

Profiling nuisance gives $\|R_N(y - F\boldsymbol{\alpha})\|_2^2$. Decompose this residual orthogonally into $\mathcal{D}$ and $\mathrm{range}(R_N) \cap \mathcal{D}^{\perp}$. The first component is the squared norm in (46); because $\overline{F}\boldsymbol{\alpha} \in \mathcal{D}$, the second is $C(y) = \|(I - U_{\mathcal{D}}U_{\mathcal{D}}^{\mathsf{T}})R_N y\|_2^2$ and contains no $\boldsymbol{\alpha}$. Reparameterizing nuisance without changing $\mathrm{range}(A)$ leaves $AA^{\dagger}$ and therefore every stated object unchanged. □

Thus $d_{\mathcal{D}}$ is the intrinsic dimension of the *linear profiled* event experiment. It is not a nonlinear global observability count and need not equal the number of PMUs. If candidates use different nuisance spans, or if nuisance has an informative proper prior, the corresponding candidate-specific profile or marginal likelihood must be retained; the common projector above is then insufficient.

Theorem 7 (Signed first-order support-to-support separation): For true support S with fixed severity direction $\boldsymbol{\alpha}_S$ and competing support C with freely fitted local amplitudes, define

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_S) = \min_{u,\boldsymbol{\beta}_C} \|F_S\boldsymbol{\alpha}_S - Au - F_C\boldsymbol{\beta}_C\|_2^2. \tag{47}$$

Then

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_S) = \| \left(I - P_{\overline{F}_C}\right) \overline{F}_S\boldsymbol{\alpha}_S\|_2^2, \tag{48}$$

$$= \boldsymbol{\alpha}_S^{\mathsf{T}} K_{S|C}\boldsymbol{\alpha}_S, \tag{49}$$

$$K_{S|C} = \overline{F}_S^{\mathsf{T}} \left(I - P_{\overline{F}_C}\right) \overline{F}_S \succeq \mathbf{0}. \tag{50}$$

Here $P_{\overline{F}_C} = \overline{F}_C\overline{F}_C^{\dagger}$, so rank deficiency in either nuisance or event coordinates does not change the projected trajectory.

b: Proof.
For fixed $\boldsymbol{\beta}_C$, minimizing (47) over u removes the component in $\operatorname{range} A$ and leaves $\|\overline{F}_S\boldsymbol{\alpha}_S - \overline{F}_C\boldsymbol{\beta}_C\|_2^2$. A second least-squares projection gives (48). Symmetry and idempotence of the orthogonal projector give (49) and its positive-semidefinite order. □

The theorem exposes two distinct weak-event mechanisms. For nonzero residualized single-source signatures $\overline{f}_i$ and $\overline{f}_j$,

$$\gamma^2_{i\to j}(a_i) = a_i^2\|\overline{f}_i\|_2^2(1-\rho_{ij}^2), \qquad \rho_{ij} = \frac{\overline{f}_i^{\mathsf{T}}\overline{f}_j}{\|\overline{f}_i\|_2\|\overline{f}_j\|_2}. \tag{51}$$

Small $\|\overline{f}_i\|$ is event–nuisance invisibility; large $|\rho_{ij}|$ is event–event aliasing after nuisance removal.

For general supports, write $H = S \cap C$, $B = S \setminus C$, and $D = C \setminus S$. The shared amplitudes can be refitted, so

$$\gamma^2_{S\to C} = \| \left(I - P_{[A,F_H,F_D]}\right) F_B\boldsymbol{\alpha}_B\|_2^2. \tag{52}$$

Thus the shared true component cancels exactly at first order for signed amplitudes or an interior parameter point. In particular, for $S = \{i,j\}$ and $C = \{i,k\}$ the relaxed margin is independent of the amplitude at i. Enlarging a competitor can only reduce the margin because its explanatory subspace grows.

The exclusive-support quotient removes the zero directions introduced by a literal support intersection. Define

$$\begin{aligned} Z_H &= [A, F_H], \quad & R_H &= I - Z_H Z_H^{\dagger}, \\ \widetilde{F}_B &= R_H F_B, \quad & \widetilde{F}_D &= R_H F_D. \end{aligned} \tag{53}$$

Then

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_B) = \min_{\boldsymbol{\beta}_D} \|\widetilde{F}_B\boldsymbol{\alpha}_B - \widetilde{F}_D\boldsymbol{\beta}_D\|_2^2$$

$$= \boldsymbol{\alpha}_B^{\mathsf{T}} K_{B|D;H}\boldsymbol{\alpha}_B, \tag{54}$$

$$K_{B|D;H} = \widetilde{F}_B^{\mathsf{T}}(I - P_{\widetilde{F}_D})\widetilde{F}_B \succeq \mathbf{0}.$$

The first equality follows by minimizing first over the common span $\operatorname{range} Z_H$ and then over $\boldsymbol{\beta}_D$. Moreover, $K_{B|D;H} \succ \mathbf{0}$ if and only if

$$\ker \widetilde{F}_B = \{\mathbf{0}\}, \qquad \operatorname{range} \widetilde{F}_B \cap \operatorname{range} \widetilde{F}_D = \{\mathbf{0}\}. \tag{55}$$

Indeed, a zero quadratic form means that $\widetilde{F}_B\boldsymbol{\alpha}_B$ lies in the competing range; the two conditions force $\boldsymbol{\alpha}_B = \mathbf{0}$, and the converse is immediate. Under the coefficient normalization $\|\boldsymbol{\alpha}_B\|_2 = 1$, Rayleigh–Ritz gives the worst relaxed direction and margin as

$$\min_{\|\boldsymbol{\alpha}_B\|_2=1} \gamma^2_{S\to C} = \lambda_{\min}(K_{B|D;H}). \tag{56}$$

This value depends on the physical scaling of the amplitude coordinates.

For a scale-free comparison, let Q_B, Q_D be orthonormal bases of the two conditional ranges. If $\theta_{\min}$ is their smallest principal angle,

$$\lambda_{\min}\left[Q_B^{\mathsf{T}}(I - Q_DQ_D^{\mathsf{T}})Q_B\right] = \sin^2\theta_{\min}. \tag{57}$$

The angle must be computed after removing nuisance and shared directions; otherwise every shared event creates a trivial zero angle. If the residual ambient dimension is $q_H = m - \operatorname{rank} Z_H$, then $r_B + r_D > q_H$ forces a nonzero intersection between the two conditional *spans*. This is a span-relaxed dimensional obstruction, not yet a one-sided physical ambiguity.

If $S \subset C$ and active amplitudes may equal zero, the competitor contains the true response and $\gamma_{S\to C} = 0$. If zero is excluded but belongs to the amplitude-domain closure, the infimum remains zero. Pure best-fit geometry therefore cannot select cardinality among nested support models. The marginal likelihood in Supplementary Appendix E supplies the required amplitude-volume and support-prior terms.

F. PHYSICAL TANGENT-CONE SEPARATION

Let $\mathcal{A}_C$ be the physical amplitude domain of a competing event model at the nominal point, and let $T_C = T_{\mathcal{A}_C}(\mathbf{0})$ denote its Bouligand tangent cone. Signed load changes give $T_C = \mathbb{R}^{|C|}$; one-sided trip, shedding, outage-continuation, and fault-severity coordinates give nonnegative rays. Missing PMU data are a discrete measurement-mode change and are not forced into this continuous construction.

Theorem 8 (Constrained first-order separation): Suppose a competing trajectory family has a local C^2 effective chart ϕ with $\phi(\mathbf{0}) = y_0$, full-column-rank derivative J, and local parameter cone $K = \mathbb{R}^p \times T_C$, where T_C is closed and polyhedral. Assume no distant branch approaches y_0. For $y_\star(\epsilon) = y_0 + \epsilon b + o(\epsilon)$,

$$d_C^2(\epsilon) = \epsilon^2 \operatorname{dist}^2(b, JK) + o(\epsilon^2). \tag{58}$$

a: Proof.

For every $v \in K$, conicity gives $\phi(\epsilon v) = y_0 + \epsilon J v + o(\epsilon)$, which yields the upper bound after taking the infimum. Conversely, an $o(\epsilon)$-optimal local coordinate u_ϵ satisfies $\|\phi(u_\epsilon) - y_0\| = O(\epsilon)$. Full column rank and differentiability imply $\|u_\epsilon\| = O(\epsilon)$. Every convergent subsequence of u_ϵ/ϵ has a limit in the closed cone K, and division of the residual by ϵ gives the matching lower bound. The isolated-branch assumption prevents a nonlocal chart from attaining a smaller leading distance. □

With A the whitened unconstrained nuisance tangent and F_C the event tangent, nuisance profiling is still exact:

$$\gamma^2_{\text{phys}} = \min_{\beta_C \in T_C} \|R_N(b - F_C\beta_C)\|_2^2 = \text{dist}^2(R_N b, R_N F_C T_C). \tag{59}$$

The observable projection is unique even when the coefficient vector is not. Let $\mathcal{C} = R_N F_C T_C$ and $\mathcal{L} = \text{span}\,\mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{L}$,

$$\gamma_{\text{phys}} \geq \gamma_{\text{span}}, \quad \gamma_{\text{phys}} = \gamma_{\text{span}} \iff \Pi_{\mathcal{L}}(R_N b) \in \mathcal{C}. \tag{60}$$

Thus a positive span margin certifies physical separation, whereas a zero span margin does not prove one-sided ambiguity.

For $T_C = \mathbb{R}_+^{|C|}$, the optimum is a nonnegative least-squares projection. With residual $r = R_N(b - F_C\beta_C)$, the KKT conditions are

$$\beta_C \geq 0, \quad (R_N F_C)^\top r \leq 0, \quad \beta_j (R_N f_j)^\top r = 0. \tag{61}$$

For one true signature x and one admissible competing ray d,

$$\gamma^2_{\text{ray}} = \|x\|_2^2 - \frac{[d^\top x]_+^2}{\|d\|_2^2}, \qquad [v]_+ = \max(0, v). \tag{62}$$

An antiparallel signature can therefore have zero span margin and a strictly positive physical margin.

For a shared one-sided support H, the exact difference coordinate satisfies $\delta_H \in T_H - \alpha_H$, which need not be a subspace. Hence the physical margin can depend on the shared true amplitude; the unconditional cancellation statement applies only when the relaxed solution is feasible. Competitor enrichment still cannot increase the margin whenever the old feasible set is contained in the new one. Likewise, a nested superset still has zero margin when it can match the shared amplitudes and set every added amplitude to zero.

For nontrivial closed cones $\mathcal{C}_S, \mathcal{C}_C$, the directed scale-free separation

$$\delta(\mathcal{C}_S \to \mathcal{C}_C) = \min_{\substack{x \in \mathcal{C}_S \\ \|x\|_2 = 1}} \text{dist}(x, \mathcal{C}_C) \tag{63}$$

vanishes exactly when the cones share a nonzero vector. It reduces to $\sin\theta_{\min}$ for subspaces, but need not be symmetric. Consequently, the classical signed-sparse condition $\text{spark}(\overline{F}) > 2K$ remains sufficient under one-sided constraints, while its failure need not imply a physically feasible ambiguity.

TABLE 3. Deterministic checks of conditional and one-sided geometry. These toy systems are not IEEE 39-bus evidence.

Check	Recomputed outcome
Conditional quotient	full–quotient error 0.0e+00 (pass).
Worst coefficient direction	eigenvalue–grid error 1.9e-08 (pass).
Conditional angle	naive 0.0e+00 deg; conditional 18.689 deg (pass).
Ray versus span	squared margins 0.0 versus 1.0 (pass).
One-sided shared source	squared-margin sequence 5.0, 2.0, 1.0, 1.0 (pass).
Cone enrichment	2000 trials; max violation 3.6e-15 (pass).
Nested cone	squared margin 2.4e-31 (pass).

For a multi-source support S, the Fisher information conditional on that known support is

$$I_S(a) = J_S(a)^\top G J_S(a), \quad J_S = \frac{\partial \mu_S}{\partial a_S}. \tag{64}$$

Positive definiteness of I_S permits local amplitude estimation conditional on S; it does not exclude another support with the same mean.

Unknown onset has a separate degeneracy. For $\mu(a, \tau) = a d(\tau)$ and fixed covariance,

$$I(a, \tau) = \begin{bmatrix} d^\top \Sigma^{-1} d & a d^\top \Sigma^{-1} \dot{d} \\ a \dot{d}^\top \Sigma^{-1} d & a^2 \dot{d}^\top \Sigma^{-1} \dot{d} \end{bmatrix}. \tag{65}$$

Profiling amplitude by a Schur complement gives

$$I_{\tau|a} = a^2 \left[\dot{d}^\top \Sigma^{-1} \dot{d} - \frac{(d^\top \Sigma^{-1} \dot{d})^2}{d^\top \Sigma^{-1} d} \right]. \tag{66}$$

At $a = 0$, onset is unidentifiable. If $\dot{d}$ is proportional to d, delay and gain are locally confounded even for nonzero amplitude. The evaluated bank avoids this unresolved problem by fixing onset.

G. CANDIDATE INDISTINGUISHABILITY AND SPARSE INJECTIVITY

For candidate h and fixed integrity support S, define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{range}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \tag{67}$$

The profiled margin in the main paper is half the squared distance between two such sets under R_H^{-1}.

Theorem 9 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the candidate manifolds intersect. At any intersection, the two candidates induce the same Gaussian law; no decision rule can distinguish them uniformly over the declared nuisance set.

a: Proof.

Zero distance between closed affine sets means that there exist nuisance vectors for which their means coincide. The covariance is common by the stated contract, so the two conditional probability measures are identical. For any decision region, the probabilities of choosing i and j under this common law sum to one; at least one conditional error is therefore at least one half. □

For two fixed simple candidates with squared Mahalanobis distance $\Delta^2 = 2D_{ij,S}$ and equal priors, projection onto their mean difference reduces the optimal test to one dimension, giving

$$P_e^\star = \Phi(-\Delta/2) = \Phi(-\sqrt{D_{ij,S}/2}). \tag{68}$$

This formula does not apply directly to amplitude-marginalized support mixtures.

For the local persistent-event model $Y_T = \mathcal{H}_T a + \varepsilon_T$, define the physically admissible difference set

$$\mathcal{D}_K = \{a - b : a, b \text{ admissible and } K\text{-sparse}\}. \tag{69}$$

Theorem 10 (Finite-horizon sparse event uniqueness): Every admissible K-sparse event is uniquely determined in the noise-free local model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{D}_K = \{\mathbf{0}\}. \tag{70}$$

For a signed neighborhood in each support subspace, $\mathcal{D}_K = \mathcal{A}_{2K}$ locally, recovering the condition in the main paper.

b: Proof.

If two admissible events produce the same output, their nonzero difference belongs to both sets in (70). Conversely, any nonzero vector in the intersection is, by definition of $\mathcal{D}_K$, the difference of two admissible K-sparse events that produce the same output. □

Unknown initial-state nuisance can be removed locally by whitening and projection:

$$\widetilde{\mathcal{H}}_T = \left(I - P_{\mathrm{range}(W_T\mathcal{O}_T)}\right) W_T \mathcal{H}_T, \qquad W_T^\mathsf{T} W_T = R_T^{-1}. \tag{71}$$

Theorem 10 then applies with $\widetilde{\mathcal{H}}_T$. Define the restricted margin

$$\gamma_{2K}(T) = \min_{1 \le |U| \le 2K} \sigma_{\min}(\widetilde{\mathcal{H}}_{T,U}). \tag{72}$$

If two K-sparse explanations both fit data within whitened residual ϵ, then

$$\|a - b\|_2 \le \frac{2\epsilon}{\gamma_{2K}(T)}. \tag{73}$$

Indeed, the triangle inequality bounds $\|\widetilde{\mathcal{H}}_T(a-b)\|_2$ by 2ϵ, while the definition of γ_{2K} gives the reverse lower bound. For the evaluated $K \le 2$ bank, the relevant union size is four.

H. INFORMATION CANNOT DECREASE WHEN VALID MEASUREMENTS ARE ADDED

For a true trajectory at severity a and a competing physical family with parameter $\boldsymbol{\eta} \in \Theta_C$, define the exact profiled prefix distance

$$d_T^2(a) = \inf_{\boldsymbol{\eta} \in \Theta_C} r_T(a, \boldsymbol{\eta})^\mathsf{T} \boldsymbol{\Sigma}_T^{-1} r_T(a, \boldsymbol{\eta}), \tag{74}$$

where every $\boldsymbol{\Sigma}_T$ is a principal prefix of one positive-definite process covariance and the physical competitor domain does not change with T. Partition the difference between candidate means into old and added blocks, $\boldsymbol{\delta} = \mathrm{col}(\boldsymbol{\delta}_1, \boldsymbol{\delta}_2)$, with joint covariance $\boldsymbol{\Sigma} \succ \mathbf{0}$. Block Gaussian elimination gives

$$\boldsymbol{\delta}^\mathsf{T} \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta} = \boldsymbol{\delta}_1^\mathsf{T} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\delta}_1 + \widetilde{\boldsymbol{\delta}}_2^\mathsf{T} \boldsymbol{\Sigma}_{2|1}^{-1} \widetilde{\boldsymbol{\delta}}_2, \tag{75}$$

$$\widetilde{\boldsymbol{\delta}}_2 = \boldsymbol{\delta}_2 - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\delta}_1, \tag{76}$$

$$\boldsymbol{\Sigma}_{2|1} = \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12}. \tag{77}$$

Theorem 11 (Prefix monotonicity and temporal rescue): Under the prefix and fixed-domain contract above,

$$d_{T+1}^2(a) \ge d_T^2(a). \tag{78}$$

If both minima are attained, equality holds if and only if an element of $\operatorname{Argmin} d_T^2$ also has zero conditional innovation residual in the added block. Suppose further that, locally,

$$d_T^2(a) = c_T a^{2k_T} + o(a^{2k_T}), \qquad c_T > 0. \tag{79}$$

Then $k_{T+1} \le k_T$; if the order remains k, then $c_{T+1} \ge c_T$.

a: Proof.

The conditional covariance in (75) is positive definite, so its second quadratic form is nonnegative for every fixed $\boldsymbol{\eta}$. Taking the infimum over the same Θ_C proves (78). If the minima are attained, equality of the infima forces an extended-horizon minimizer to minimize the prefix term and to make the innovation term zero; the converse is immediate. If $k_{T+1} > k_T$, the ratio $d_{T+1}^2(a)/d_T^2(a)$ tends to zero as $a \to 0$, contradicting (78). When both orders equal k, division by a^{2k} and passage to the limit yield $c_{T+1} \ge c_T$. □

The theorem permits a longer prefix to reveal a first-order direction that was absorbed at an earlier horizon, or to reveal a higher-order obstruction that was previously invisible. It does not say that a concrete estimator improves: changing its normalization, nuisance set, covariance model, or optimization breaks the prefix contract. In particular, the archived RAWSIM39 calibration renormalized each event–signal–horizon block to unit weight and is not a prefix likelihood for this theorem.

If whitened model errors under two supports satisfy $\|We_S\| \le \epsilon_S$ and $\|We_R\| \le \epsilon_R$, the reverse triangle inequality gives the robust separation bound

$$d_{\mathrm{true}} \ge [d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+, \qquad \mathcal{J}_{\mathrm{true}} \ge \tfrac{1}{2}[d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+^2. \tag{80}$$

This is the required bridge for future simulator-shift claims; it is not evaluated by the current nominal load bank.

I. TEMPORAL INFORMATION UNDER THE FROZEN AR(1) CONTRACT

For true support S, competing support R, true amplitude a, and competitor amplitude b, let $\boldsymbol{\delta}_t(b) = \boldsymbol{\mu}_{S,t}(a) - \boldsymbol{\mu}_{R,t}(b)$. The frozen temporal covariance is

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}, \qquad [R_T]_{ij} = \rho^{|i-j|}, \quad |\rho| < 1. \tag{81}$$

For fixed b, its squared Mahalanobis separation has the innovation form

$$q_T(b) = \|\boldsymbol{\delta}_1(b)\|^2_{\boldsymbol{\Omega}^{-1}} + \frac{1}{1-\rho^2}\sum_{t=2}^{T}\|\boldsymbol{\delta}_t(b) - \rho\boldsymbol{\delta}_{t-1}(b)\|^2_{\boldsymbol{\Omega}^{-1}}. \quad (82)$$

Define the profiled simple-model separation

$$\mathcal{J}_{S\to R}(T) = \tfrac{1}{2}\inf_{b\in\mathcal{A}_R} q_T(b). \quad (83)$$

It equals the minimum KL divergence over fixed-amplitude simple Gaussian competitors. It is not the KL divergence or Bayes factor of the amplitude-mixture support model.

Theorem 12 (Persistent AR(1) information rate): If the amplitude domain is compact and fixed and $\boldsymbol{\delta}_t(b)$ converges uniformly to $\boldsymbol{\delta}_\infty(b)$, then

$$\lim_{T\to\infty}\frac{2\mathcal{J}_{S\to R}(T)}{T} = \frac{1-\rho}{1+\rho}\inf_{b\in\mathcal{A}_R}\|\boldsymbol{\delta}_\infty(b)\|^2_{\boldsymbol{\Omega}^{-1}}. \quad (84)$$

a: Proof.

Under uniform convergence, $\boldsymbol{\delta}_t - \rho\boldsymbol{\delta}_{t-1}$ converges uniformly to $(1-\rho)\boldsymbol{\delta}_\infty$. Divide (82) by T; the first-frame term vanishes and the Cesàro mean converges uniformly to $(1-\rho)^2(1-\rho^2)^{-1}\|\boldsymbol{\delta}_\infty\|^2_{\boldsymbol{\Omega}^{-1}}$. Compactness and uniform convergence permit interchange of infimum and limit, giving (84). □

Monotonicity of $\mathcal{J}_{S\to R}(T)$ follows independently from Theorem 11; the AR(1) calculation supplies the explicit limiting rate.

If the profiled equilibrium distance is positive, separation grows as $\Theta(T)$. If an admissible competitor matches equilibrium and its innovation differences are square summable, evaluating (83) at that competitor gives a finite upper bound; monotonicity then implies convergence to a finite ceiling. Exact trajectory overlap gives zero separation for every horizon.

For a true pair $\{i,j\}$, a competing pair $\{i,k\}$ can refit the shared source. With whitened equilibrium columns $\bar{h}_i, \bar{h}_j, \bar{h}_k$,

$$a_j^2\|(I - P_{i,k})\bar{h}_j\|_2^2 \le a_j^2\|(I-P_i)\bar{h}_j\|_2^2. \quad (85)$$

The inequality follows from $\mathrm{span}(\bar{h}_i) \subseteq \mathrm{span}(\bar{h}_i, \bar{h}_k)$. It formalizes why identifying which weak second source occurred cannot be easier than rejecting the shared single-source explanation.

J. HIGHER-ORDER CONTACT AND LOCAL DETECTION SCALE

First-order Fisher information can vanish while nonlinear separation remains. In whitened coordinates, let the no-event nuisance manifold and event curve be

$$F_0(b) = Nb + \tfrac{1}{2}H_0[b,b] + O(\|b\|^3), \quad (86)$$
$$c(a) = ad + a^2 q + O(a^3). \quad (87)$$

Theorem 13 (Relative first- and second-order separation): Assume the nuisance manifold has constant rank and use a regular chart after removing redundant coordinates. Define the profiled first-order coefficient

$$\gamma_1^2 = \|P_\perp d\|_2^2, \qquad P_\perp = I - (WJ_c)(WJ_c)^\dagger, \quad (88)$$

where J_c is the nuisance tangent before whitening and d is the already whitened event tangent in (87). If $\gamma_1 > 0$, the local squared distance to the nuisance manifold is

$$\Delta^2(a) = a^2\gamma_1^2 + O(a^3). \quad (89)$$

If $\gamma_1 = 0$, so the whitened event tangent lies in the nuisance tangent space, choose the regular-chart vector v satisfying $Nv = d$ and define

$$\boldsymbol{\kappa} = P_\perp\left(q - \tfrac{1}{2}H_0[v,v]\right). \quad (90)$$

Then

$$\Delta^2(a) = a^4\gamma_2^2 + O(a^5), \qquad \gamma_2^2 = \|\boldsymbol{\kappa}\|_2^2. \quad (91)$$

a: Proof.

The local minimizer satisfies $DF_0(b)^\mathsf{T}[F_0(b) - c(a)] = \mathbf{0}$. At the origin, its derivative with respect to the regular chart coordinate is $N^\mathsf{T}N \succ \mathbf{0}$, so the implicit-function theorem gives a smooth minimizing branch. Write $b_\star(a) = av + a^2 w + O(a^3)$. At first order, $N^\mathsf{T}(Nv - d) = \mathbf{0}$, yielding residual $aP_\perp d + O(a^2)$ and (89). If $d = Nv$, the first-order residual vanishes. At second order, minimization projects $q - \frac{1}{2}H_0[v,v]$ onto $\mathrm{range}\,N$; the remaining normal component is (90). Squaring the residual gives (91). The projector is unchanged by any full-rank reparameterization of the same nuisance tangent space, so γ_1 is a property of the local manifolds and covariance metric rather than of a particular nuisance coordinate basis. □

The same constant-rank qualification makes γ_2 intrinsic. Redundant original parameters may give several representatives of the same observable tangent, but their normal acceleration equals the second fundamental form of the same manifold branch. If $\gamma_2 = 0$, the conclusion is only $\Delta^2(a) = o(a^4)$ under the corresponding little-o expansion; an a^6 law or exact overlap requires a separate third-order analysis.

K. OPERATIONAL ERROR AND SUPPORT-RECOVERY LIMITS

For two simple hypotheses with a shared covariance, $P_i = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, define

$$d^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0). \quad (92)$$

The equal-prior log likelihood ratio is Gaussian with mean $\mp d^2/2$ and variance d^2 under P_0 and P_1, respectively. Hence

$$P_e^\star = \Phi(-d/2), \quad D_{\mathrm{KL}}(P_1\|P_0) = d^2/2, \quad C_{\mathrm{Ch}} = d^2/8. \quad (93)$$

The last identity gives $P_e^\star \le \frac{1}{2}\exp(-d^2/8)$. If a common affine nuisance is $A\boldsymbol{\eta}$, orthogonal projection after whitening removes it exactly and replaces d by the quotient margin.

Thus $P_e^\star = \Phi(-\gamma/2)$ is exact for that quotient Gaussian experiment. It is not an exact formula for two nonlinear amplitude mixtures with different priors or dimensions.

Suppose the first nonzero contact order at horizon T is k and

$$d_T^2(a) = T\mathcal{I}_k|a|^{2k}[1+o(1)], \qquad \mathcal{I}_k > 0. \tag{94}$$

For target simple-Gaussian error $\alpha < 1/2$, inversion of (93) yields

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}}\right]^{1/k}. \tag{95}$$

Equivalently, if $a_T = cT^{-\beta}$, the optimal binary error tends to one half for $\beta > 1/(2k)$, to a nontrivial constant at equality, and to zero for $\beta < 1/(2k)$. This gives the $T^{-1/2}$ and $T^{-1/4}$ boundaries for first- and second-order separation.

Now let $\Theta = \bigcup_{S\in\mathfrak{S}} \Theta_S$ and define the minimax exact-support risk

$$R^\star = \inf_{\widehat{S}} \sup_{\theta\in\Theta} \Pr_\theta\{\widehat{S} \neq S(\theta)\}. \tag{96}$$

Theorem 14 (Binary and minimum-severity limits): For every pair $\theta_i \in \Theta_{S_i}$ with $S_0 \neq S_1$,

$$\begin{aligned} R^\star &\geq \frac{1}{2}[1 - \mathrm{TV}(P_{\theta_0}, P_{\theta_1})] \\ &\geq \frac{1}{2}\left[1 - \sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(P_{\theta_0}\|P_{\theta_1})}\right]_+. \end{aligned} \tag{97}$$

If $S^+ = S \cup \{j\}$ and $P_{S^+,a_j} \to P_S$ in total variation as $a_j \downarrow 0$, then uniform exact-support recovery over all $a_j > 0$ has minimax risk at least one half.

a: Proof.

Restriction of any support rule to a two-point subset cannot reduce its worst-case risk; the optimal equal-prior binary risk is $[1 - \mathrm{TV}(P,Q)]/2$. Pinsker's inequality gives the second bound. For the nested pair, the binary risk tends to one half as the active amplitude tends to zero, so the supremum over positive amplitudes is at least that limit. □

The theorem requires either a declared $|a_j| \geq a_{\min} > 0$ contract or a decision that may abstain near absence. It also distinguishes a source-ranking error from an impossible exact-cardinality decision.

For M equiprobable supports, Fano's inequality gives

$$P_e \geq 1 - \frac{I(V;Y) + \log 2}{\log M}. \tag{98}$$

With $\overline{P} = M^{-1}\sum_j P_j$, convexity of KL in its second argument implies

$$\begin{aligned} I(V;Y) &= \frac{1}{M}\sum_i D_{\mathrm{KL}}(P_i\|\overline{P}) \\ &\leq \frac{1}{M^2}\sum_{i,j} D_{\mathrm{KL}}(P_i\|P_j) = \overline{D}_{\mathrm{KL}}. \end{aligned} \tag{99}$$

Therefore a necessary condition for $P_e \leq \alpha$ is

$$\overline{D}_{\mathrm{KL}} \geq (1-\alpha)\log M - \log 2. \tag{100}$$

If $M = \binom{N}{K}$ and $\overline{D}_{\mathrm{KL}}(T) \leq T\overline{\mathcal{I}}_k a^{2k}/2$, then necessarily

$$T \geq \frac{2[(1-\alpha)\log\binom{N}{K} - \log 2]}{\overline{\mathcal{I}}_k a^{2k}}. \tag{101}$$

This is an impossibility certificate, not a sufficient sample-size guarantee [17], [18]. For sparse $K \ll N$, it has the familiar scaling $T \gtrsim K\log(N/K)/(\overline{\mathcal{I}}_k a^{2k})$, now expressed through the physical Hybrid-DAE information rate.

L. DIAGNOSABILITY-PRESERVING DISCREPANCY COVARIANCE

A residual correction can improve fit while making physical events harder to distinguish. The following calculation makes that tradeoff explicit. Work after whitening and removal of common nuisance. Let $Q_U \in \mathbb{R}^{m\times r}$ be an orthonormal basis for the discriminant directions that must be protected, and let $Q_D \in \mathbb{R}^{m\times q}$ be an orthonormal discrepancy basis. The former should be frozen from critical candidate pairs; the latter must be estimated from mismatch-only records so that unlabeled events are not absorbed as model error.

Theorem 15 (Discrepancy retention and finite intensity): Let $P_D = Q_D Q_D^\mathsf{T}$ and model the whitened residual with covariance

$$C_s = I + sP_D, \qquad s = \tau^2 \geq 0. \tag{102}$$

Then $C_s^{-1} = I - \alpha P_D$, where $\alpha = s/(1+s)$, and the worst information fraction retained over the protected subspace is

$$\rho(s, Q_D) = 1 - \alpha\,\sigma_{\max}^2(Q_U^\mathsf{T}Q_D). \tag{103}$$

Set $c^2 = \sigma_{\max}^2(Q_U^\mathsf{T}Q_D)$ and $\delta = 1 - \rho_{\min}$. The constraint $\rho \geq \rho_{\min}$ imposes no finite upper bound on s when $c^2 \leq \delta$; otherwise it requires

$$s \leq s_{\max} = \frac{\delta}{c^2 - \delta}. \tag{104}$$

If the true residual covariance is $I + C_m$ and $E_D = \mathrm{tr}(Q_D^\mathsf{T} C_m Q_D)$, the expected Gaussian negative twice-log-likelihood relative to $s = 0$, up to constants, is

$$\Delta\mathcal{L}(s) = q\log(1+s) - \frac{s}{1+s}(q + E_D). \tag{105}$$

For fixed Q_D, its constrained minimizer is finite:

$$s^\star = \min\{E_D/q, s_{\max}\}, \tag{106}$$

with $s_{\max} = +\infty$ in the first geometric case.

a: Proof.

The inverse follows from $P_D^2 = P_D$. For $x = Q_U a$, the information ratio is the Rayleigh quotient of $I - \alpha Q_U^\mathsf{T} P_D Q_U$; its smallest eigenvalue is (103) because the singular values of $Q_U^\mathsf{T}Q_D$ are the principal-angle cosines. Solving $\alpha c^2 \leq \delta$ gives (104). Finally, $\log|C_s| = q\log(1+s)$ and taking the expectation of $y^\mathsf{T}C_s^{-1}y$ under $\mathrm{Cov}(y) = I + C_m$ gives (105). Its derivative is $(qs - E_D)/(1+s)^2$, which gives (106). □

Under strict no-loss protection, $\rho_{\min} = 1$ and $s > 0$ require $\mathrm{range}(Q_D) \subseteq \mathrm{range}(Q_U)^\perp$. For fixed rank, the basis

TABLE 4. Deterministic checks for the discrepancy-retention identities. These are algebraic tests, not power-system experiments.

Check	Error	Tolerance	Status
Low-rank covariance inverse	4.53e-16	1.0e-12	PASS
Principal-angle retention	2.22e-16	1.0e-12	PASS
Retention variance cap	0.00e+00	1.0e-12	PASS
Finite full-likelihood intensity	5.50e-09	1.0e-07	PASS
Safe projected PCA	1.78e-15	1.0e-10	PASS
Protected-distance floor	1.11e-16	1.0e-12	PASS

that captures the most mismatch energy is therefore the top-q eigenspace of $P_{U^\perp} C_m P_{U^\perp}$, provided that the orthogonal complement has dimension at least q. This is a safe projected-PCA baseline, not a learned-estimator result.

Maximizing only αE_D omits both the covariance determinant and the whitened measurement-noise contribution. It therefore drives s toward an infinite-variance limit whenever geometry permits. That utility is a valid description of reduced quadratic mismatch penalty, but it is not a normalized Bayesian tuning objective. The general joint orientation problem for $q > 1$ is an offline nonconvex design; spectral, semidefinite, or moment relaxations belong in a separate optimization study unless they are executed on the frozen event and mismatch dictionaries.

M. APPLIED GATE TAXONOMY FROM AN ARCHIVED ANDES CHECKPOINT

The frozen RAWSIM39 V1.2 calibration ledger provides an applied boundary check on the theory. It contains complete first-order screening and rank-selection decisions for 18 plant/event directions, but not the Jacobian and higher-order arrays needed to evaluate γ_2 or γ_3. Table 5 therefore classifies only what the recorded gates establish. In particular, failure of the curvature gate is not relabeled as second-order identifiability, and a small orthogonal fraction is not relabeled as exact redundancy.

N. THEORY-TO-EVIDENCE BOUNDARY

Equation (89) supplies a direct falsification test: for independently generated nonlinear trajectories with decreasing $|a|$, the profiled squared distance divided by a^2 should converge to γ_1^2. That convergence study has not been run for the present ANDES bank; the retrospective distance association and the gate-only RAWSIM39 taxonomy are not substitutes for it.

If n independent Gaussian replications have local distance $\Delta(a) = |a|^r \|k_r\| + O(|a|^{r+1})$, their KL divergence to the closest null is $n\Delta^2(a)/2$. Pinsker's inequality bounds the power of any level-α test by

$$\mathrm{Power}_a \le \alpha + \min\{1, \sqrt{n}\Delta(a)/2\}. \tag{107}$$

Thus $n|a_n|^{2r} \to 0$ prevents power from separating from size. The reference scales are $n^{-1/2}$ for first-order contact and $n^{-1/4}$ for second-order contact; singular-information asymptotics are established more broadly [19]. Correlated frames in one PMU window are not independent replications and must be handled through their joint covariance.

All audited load directions in the present bank have nonzero first-order separation. The second-order result is therefore a boundary theorem, not the empirical explanation for weak-pair failure.

TABLE 5. Conservative taxonomy of the frozen RAWSIM39 V1.2 calibration checkpoint. The gate sequence was 18 candidates, 10 individually eligible ($\kappa = 722.91$), eight rank-selected ($\kappa = 7.83$), and seven released after one 6.15σ local-domain rejection.

Category	Count	Directions	Evidence boundary
Regular local	7	plant_inertia_scale, plant_governor_droop_scale, generation_delta_p_pu, generation_delta_q_pu, generation_amplitude, generation_peak_s, load_delta_p_fraction	Released only under the local restricted posterior
Weak first-order	6	plant_damping, plant_governor_lag_scale, plant_turbine_time_scale, plant_avr_gain_scale, plant_avr_time_scale, plant_subtransient_x_scale	Low information; order and temporal rescue unresolved
Curvature unresolved	2	fault_duration_s, line_duration_s	Linear surrogate failed; not evidence of order 2
Near-redundant first-order	2	generation_tau_s, generation_residual	Small new orthogonal fraction; exact redundancy unproved
Local-domain failure	1	fault_reactance_pu	MAP left the declared local prior domain

TABLE 6. Recomputed checks of higher-order evidence and information limits. Controlled systems are not IEEE 39-bus performance evidence.

Check	Recomputed outcome
Evidence classes	slopes 0.9994, 0.4951, and 0.0003 for regular, order-2, and redundant additions (pass).
Prefix monotonicity	500 correlated-noise trials; max violation 0.0e+00 (pass).
Temporal rescue	first-order $\gamma_1^2(2) = 0.452724$; second-order relative error 1.0e-04 (pass).
Gaussian error map	quadrature error 5.6e-17 (pass).
Fano lower bound	$M = 8$ bounds 0.5993/0.3974; no Monte Carlo violation (pass).

TABLE 7. Deterministic checks of the local separation identities. These toy systems are not IEEE 39-bus evidence.

Check	Prediction and recomputed outcome
Regular first-order DAE	slope 2 and $\gamma_1^2 = 0.4527244641$; 2.000000; relative error 3.8e-12 (pass).
Shared-tangent second order	slope 4 and $\gamma_2^2 = 2.89$; 4.000000; coefficient 2.890000 (pass).
Rank-deficient nuisance	unique projected fit and $\gamma_1^2 = 0.5$; rank 2/3; max variation 1.2e-16 (pass).
Signed shared-source cancellation	margin independent of shared amplitude; squared-margin range 0.0e+00 (pass).
Competitor enrichment	$\gamma(S \to C_2) \leq \gamma(S \to C_1)$; 0.0196 versus 0.5096 (pass).
Nested support	geometric margin 0; 6.1e-35 (pass).
Principal angle	unit margin $\sin\theta$; absolute error 0.0e+00 (pass).
Nuisance enlargement	profiled margin cannot increase; sequence 2.0, 1.0, 0.0 (pass).
Diagnostic quotient	profiled costs factor through $d_{\mathcal{D}}$ coordinates; rank 4/6; max error 1.1e-15 (pass).
Nuisance reparameterization	unchanged nuisance span gives the same projector; spectral error 9.5e-16 (pass).

TABLE 8. Open checks for the simulator event semantics.

Check	Evidence required for closure
Right limit	Solver-scale $\|g_+(x(\tau+\epsilon), z(\tau+\epsilon), a)\|$ after re-equilibration; the callback residual may remain $O(a)$.
Algebraic derivatives	Agreement of analytic z_1, z_2 with central differences on a decreasing symmetric amplitude grid.
Taylor order	Constraint and state-approximation slopes $1, 2, 3$, or amplitude-halving ratios $2, 4, 8$.
Event families	One declared semantics and tolerance set for load, generation, and continuous branch-admittance perturbations.
Dynamic first order	Convergence of $\Delta y/a$ to $B_{r,a}$ over the declared early-time window.
Dynamic second order	Analytic second variation within a preregistered tolerance of nonlinear trajectories; the current discrepancy is unresolved.

TABLE 9. Status of the mathematical results relative to the frozen evidence.

Result	Present evidence boundary
Consistency jump	DAE contract only; callback/right-limit logging and derivative convergence remain open.
Functional equivalence and BLUE	Exact in the local linear model; nonlinear TEST accuracy remains empirical.
Fisher, support, and cone geometry	Signed and one-sided toy identities pass; IEEE 39-bus convergence and physical cone margins remain untested.
Sparse injectivity and γ_{2K}	T120 diagnostics concern the frozen load bank, not general descriptor left-invertibility.
Temporal order and rate	Prefix and controlled correlated-noise identities pass; the frozen long-horizon load study remains diagnostic and lacks independent operating-point validation.
Relative curvature	The controlled parabola passes; curvature-dominated weak-event behavior was not observed in the frozen atlas.
Error and minimax limits	Gaussian quotient, minimum-severity, and Fano formulas pass controlled checks; no IEEE 39 pairwise oracle-error atlas has been frozen.
RAWSIM39 gate taxonomy	The 18-direction ledger supports five conservative gate categories; derivative arrays needed to assign higher-order classes are unavailable.
Discrepancy retention	Exact for the whitened low-rank covariance model and checked algebraically; the protected dictionary, mismatch covariance, learned correction, and shifted IEEE 39-bus test remain unexecuted.

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

This appendix distinguishes a state-reconstruction component from two event estimator contracts. Sections E-A–E-C specify the state recursion and the implemented batch physical-manifold Bayes estimator $\mathcal{E}_{\mathrm{B}}$. Section E-F specifies the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. Filtering, smoothing, quadrature, and optimization are numerical operations within those contracts, not additional estimator proposals.

A. BAYESIAN RECURSION AND THE STATE HIERARCHY

For a Markov state model, Bayesian filtering alternates prediction and correction [20]. More broadly, this construction treats the PMU task as a Bayesian inverse problem in which physical candidate models and latent states are compared through predictive evidence [21]:

$$p(x_k \mid y_{0:k-1}) = \int p(x_k \mid x_{k-1}) p(x_{k-1} \mid y_{0:k-1})\, dx_{k-1}, \tag{108}$$

$$p(x_k \mid y_{0:k}) = \frac{p(y_k \mid x_k) p(x_k \mid y_{0:k-1})}{\int p(y_k \mid x_k) p(x_k \mid y_{0:k-1})\, dx_k}. \tag{109}$$

The denominator in (109) is the predictive evidence. Omitting it is harmless for a state mean under one fixed model, but not when physical candidates are compared.

B0 returns the equilibrium. B1 conditions the equilibrium prior on one PMU frame:

$$K_{\mathrm{LMMSE}} = P_0 C^{\mathsf{T}} (C P_0 C^{\mathsf{T}} + R)^{-1}, \tag{110}$$

$$\widehat{\delta x}_k^{\mathrm{B1}} = K_{\mathrm{LMMSE}} (y_k - y_0). \tag{111}$$

B2 retains state and covariance between frames. For the linear Gaussian model, substituting Gaussian densities into (108)–(109) yields

$$m_k^- = A_d m_{k-1}^+, \qquad P_k^- = A_d P_{k-1}^+ A_d^{\mathsf{T}} + Q, \tag{112}$$

$$S_k = C P_k^- C^{\mathsf{T}} + R, \qquad K_k = P_k^- C^{\mathsf{T}} S_k^{-1}, \tag{113}$$

$$m_k^+ = m_k^- + K_k (y_k - C m_k^-), \qquad P_k^+ = (I - K_k C) P_k^-. \tag{114}$$

Theorem 16 (Linear-Gaussian reduction): Equations (112)–(114) are the exact posterior mean and covariance generated by the Bayesian recursion for the stated linear Gaussian model.

a: Proof.

An affine transformation of the Gaussian posterior at $k-1$, followed by addition of independent Gaussian process noise, gives (112). The pair (x_k, y_k) is jointly Gaussian with cross covariance $P_k^- C^{\mathsf{T}}$ and measurement covariance S_k. The standard conditional-Gaussian formula gives (113)–(114). Equivalently, multiplying the predictive density by the likelihood and completing the square gives posterior precision $(P_k^-)^{-1} + C^{\mathsf{T}} R^{-1} C$; Woodbury's identity produces the same covariance and mean. □

The implementation may use the Joseph covariance form

$$P_k^+ = (I - K_k C) P_k^- (I - K_k C)^{\mathsf{T}} + K_k R K_k^{\mathsf{T}} \tag{115}$$

to preserve symmetry and positive semidefiniteness numerically.

B3 uses the Rauch–Tung–Striebel backward recursion [22]:

$$J_t = P_t^+ A_d^{\mathsf{T}} (P_{t+1}^-)^{-1}, \tag{116}$$

$$m_{t|k} = m_t^+ + J_t (m_{t+1|k} - m_{t+1}^-), \tag{117}$$

$$P_{t|k} = P_t^+ + J_t (P_{t+1|k} - P_{t+1}^-) J_t^{\mathsf{T}}. \tag{118}$$

It estimates a past state using later measurements. The negative B3 result therefore diagnoses model mismatch; it does not imply that correct future information generally lacks value.

The nominal uncertainty correction rescales the rectangular voltage covariance as

$$\boldsymbol{\Sigma}_{i,k}^{\mathrm{U2}} = T \boldsymbol{\Sigma}_{i,k} T^{\mathsf{T}}, \qquad T = \mathrm{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}), \tag{119}$$

with temperatures fitted on independent calibration trajectories. The observed-only adequacy statistic is a five-frame trailing mean of

$$a_k = (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu)^{\mathsf{T}} \boldsymbol{\Sigma}_\nu^{-1} (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu). \tag{120}$$

Neither operation changes the physical state estimate.

B. CONSTRAINED PHYSICAL RECENTERING

The static recentering estimator treats active/reactive load changes and active generator changes as nuisance coordinates d. It solves

$$(\widehat{V}, \widehat{d}) = \arg\min_{V,d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{\mathrm{AC}}(V,d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2. \tag{121}$$

Development data fix $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. Because the AC constraint appears through a finite penalty rather than a normalized stochastic density, the implemented objective is a regularized physical estimator, not a fully normalized posterior.

For an exactly constrained local MAP, let $r(\boldsymbol{\chi})$ be the stacked residual, $c(\boldsymbol{\chi}) = \mathbf{0}$ the equality constraints, and $\boldsymbol{\Omega}$ the residual precision. A Gauss–Newton step solves

$$\begin{bmatrix} H & G_c^{\mathsf{T}} \\ G_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{\chi} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla J \\ -c \end{bmatrix}, \quad H = J_r^{\mathsf{T}} \boldsymbol{\Omega} J_r + H_{\mathrm{prior}}. \tag{122}$$

Theorem 17 (Feasible local covariance): If G_c has full row rank and Z spans $\ker G_c$, then $Z^\mathsf{T} H Z \succ \mathbf{0}$ makes the KKT matrix nonsingular. The local constrained Gaussian covariance is

$$P_\chi = Z(Z^\mathsf{T} H Z)^{-1} Z^\mathsf{T}. \tag{123}$$

a: Proof.

Any vector in the KKT nullspace satisfies $G_c v = \mathbf{0}$, hence $v = Zq$. Premultiplying the first block equation by v^T gives $q^\mathsf{T} Z^\mathsf{T} H Z q = 0$, so $q = \mathbf{0}$ and $v = \mathbf{0}$. Full row rank then forces the multiplier to zero. Parameterizing feasible perturbations by $\delta\boldsymbol{\chi} = Zq$ gives Gaussian precision $Z^\mathsf{T} H Z$ in q; its linear pushforward is (123). □

This covariance is local and unimodal. The reported joint NEES-like failure shows that it is not calibrated for the full shifted-plant experiment.

C. IMPLEMENTED BATCH PHYSICAL-MANIFOLD BAYES $\mathcal{E}_\mathsf{B}$

Let r be the centered T-frame PMU window. For support S with amplitudes a_S, the evaluated mean is

$$\boldsymbol{\mu}_S(a_S) = \sum_{i\in S}(a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j\in S\\ i<j}} a_i a_j Q_{ij}. \tag{124}$$

The W2 likelihood is

$$q_S(a_S) = \|r - \boldsymbol{\mu}_S(a_S)\|^2_{\boldsymbol{\Sigma}_T^{-1}}, \tag{125}$$

$$p(r \mid S, a_S) = \frac{\exp[-q_S(a_S)/2]}{(2\pi)^{n_y T/2}|\boldsymbol{\Sigma}_T|^{1/2}}, \tag{126}$$

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}. \tag{127}$$

The covariance is fitted only on normal calibration records and remains fixed across supports. The selected priors are

$$p(K = 0, 1, 2) = (0.20, 0.50, 0.30), \tag{128}$$

$$p(S \mid K) = \binom{16}{K}^{-1}, \tag{129}$$

$$a_i \mid i \in S \sim \mathcal{N}(0, \sigma_a^2), \qquad \sigma_a = 0.05. \tag{130}$$

For reference, the linear single-source model has a closed form.

Proposition 1 (Conjugate amplitude posterior and Bayes factor): Let $r = ad + \varepsilon$, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, and $a \sim \mathcal{N}(0, \sigma_a^2)$. Define $I = d^\mathsf{T}\boldsymbol{\Sigma}^{-1} d$ and $s = d^\mathsf{T}\boldsymbol{\Sigma}^{-1} r$. Then

$$v = (I + \sigma_a^{-2})^{-1}, \qquad m = vs, \tag{131}$$

$$a \mid r, S \sim \mathcal{N}(m, v), \tag{132}$$

$$\log \frac{p(r \mid S)}{p(r \mid H_0)} = \tfrac{1}{2}\frac{s^2}{I + \sigma_a^{-2}} - \tfrac{1}{2}\log(1 + \sigma_a^2 I). \tag{133}$$

a: Proof.

Terms in the joint log density that depend on a are $-\frac{1}{2}[(I + \sigma_a^{-2})a^2 - 2sa]$. Completing the square gives mean m and variance v. Integrating the resulting Gaussian contributes $\sqrt{v/\sigma_a^2}\exp(vs^2/2)$ relative to H_0; taking its logarithm gives (133). The logarithmic term is the amplitude-volume or Occam penalty. □

Corollary 2 (One-sided amplitude evidence): Under Proposition 1, replace the signed Gaussian slab by the proper half-normal prior $p(a) = 2\mathcal{N}(a; 0, \sigma_a^2)\mathbb{I}(a \geq 0)$. If $BF_\pm$ denotes the signed Bayes factor in (133), then

$$BF_+ = 2\Phi(z) BF_\pm, \qquad z = \frac{\sigma_a s}{\sqrt{1 + \sigma_a^2 I}}. \tag{134}$$

Conditional on the active model, the severity posterior is $\mathcal{N}(m, v)$ truncated to $[0, \infty)$.

b: Proof.

The fixed-amplitude likelihood ratio is $\exp(as - Ia^2/2)$. Multiplication by the half-normal density and completion of the square give twice the signed Gaussian integral restricted to $a \geq 0$. The unrestricted posterior has standardized probability $\Phi(m/\sqrt{v}) = \Phi(z)$ on that half-line, which yields (134). □

The factor $2\Phi(z)$ preserves direction: a large score with the prohibited sign is evidence against the one-sided event even though the signed factor depends on s^2. Proper slab scales are essential because Bayes factors across different support dimensions are undefined up to an arbitrary constant under an improper amplitude prior. Under the one-sided null $a = 0$, the likelihood ratio statistic satisfies

$$2\Delta\ell = [Z]_+^2, \qquad Z \sim \mathcal{N}(0, 1), \tag{135}$$

and therefore has the boundary mixture $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$, rather than the regular χ_1^2 law.

The quadratic multi-source mean is nonconjugate. Its evidence is

$$Z_S = p(r \mid S) = \int p(r \mid S, a_S) \prod_{i\in S} p(a_i)\, da_S. \tag{136}$$

With Q-node Gauss–Hermite abscissas x_q and weights w_q, the Gaussian-prior transformation $a_i = \sqrt{2}\sigma_a x_{q_i}$ gives

$$Z_S \approx \pi^{-|S|/2} \sum_{q_1=1}^{Q} \cdots \sum_{q_{|S|}=1}^{Q} \left(\prod_{j=1}^{|S|} w_{q_j}\right) p(r \mid S, \sqrt{2}\sigma_a x_q). \tag{137}$$

The implementation uses $Q = 31$ and evaluates (137) in the log domain using log-sum-exp. Independent adaptive Gauss–Kronrod integration checks representative two-dimensional cases. These checks support numerical integration under the frozen model; they do not validate the physical model itself.

D. POSTERIOR OUTPUTS AND THE CARDINALITY–LOCALIZATION DISTINCTION

The normalized support posterior is

$$p(S \mid r) = \frac{Z_S p(S)}{\sum_{R\in\mathcal{H}} Z_R p(R)}. \tag{138}$$

It induces cardinality, source-inclusion, and amplitude distributions:

$$p(K = k \mid r) = \sum_{|S|=k} p(S \mid r), \tag{139}$$

$$p(i \in S \mid r) = \sum_{S:\, i \in S} p(S \mid r), \tag{140}$$

$$p(a_i \mid r) = p(i \notin S \mid r)\delta_0 + \sum_{S:\, i \in S} p(S \mid r) p(a_i \mid S, r). \tag{141}$$

Equation (141) is a model-averaged spike-and-slab output: source absence contributes a point mass at zero.

The experimentally observed gap between cardinality and support has an exact algebraic decomposition. Let $N_k = \binom{16}{k}$, $\ell_S = \log Z_S$, and $\ell_k^\star = \max_{|S|=k} \ell_S$. Define the effective multiplicity

$$m_{\mathrm{eff},k} = \sum_{|S|=k} \exp(\ell_S - \ell_k^\star). \tag{142}$$

Proposition 2 (Exact effective-multiplicity identity): Under a uniform prior over supports of cardinality k,

$$\log p(r \mid K = k) = \ell_k^\star - \log N_k + \log m_{\mathrm{eff},k}, \tag{143}$$

$$\max_{|S|=k} p(S \mid K = k, r) = m_{\mathrm{eff},k}^{-1}. \tag{144}$$

a: Proof.

Factor $\exp(\ell_k^\star)$ from $N_k^{-1} \sum_{|S|=k} \exp(\ell_S)$ and take logarithms to obtain (143). Conditional normalization within cardinality k divides the largest evidence by the same sum, giving (144). □

Many similar pair supports can therefore increase evidence for $K = 2$ while preventing any one pair from dominating. Entropy separates the same two questions exactly because K is a deterministic function of S:

$$H(S \mid r) = H(K \mid r) + \sum_k p(K = k \mid r) H(S \mid K = k, r). \tag{145}$$

The first term measures cardinality uncertainty; the second measures localization ambiguity within each cardinality.

For set-valued output, sort supports by posterior mass and define the smallest credible set

$$\mathcal{C}_q(r) = \arg\min_{\mathcal{C}} |\mathcal{C}| \quad \text{subject to} \quad \sum_{S \in \mathcal{C}} p(S \mid r) \geq q. \tag{146}$$

Candidate-set size and coverage are more informative than an arbitrary Top-m when the posterior is multimodal. The present paper has not yet calibrated an abstention threshold or credible-set coverage prospectively.

E. STEP-BY-STEP EVALUATED BAYESIAN ALGORITHM

Table 10 records the exact order of the load-support computation. Steps marked "evaluated" produced the 137-hypothesis results; optional outputs are mathematically defined but are not promoted to validated claims.

The direct quadrature cost per record is

$$1 + GQ + \binom{G}{2} Q^2 \tag{147}$$

likelihood-node evaluations for $K \leq 2$. With $G = 16$ and $Q = 31$, this is 115,817 nodes before vectorization or reuse. The count explains why the support bank, integration order, and horizon must be frozen together when runtime is compared.

An incremental implementation of $\mathcal{E}_{\mathrm{B}}$ is computational, not a new estimator. If the exact innovations factorization includes the same initial AR(1) density, then at every frozen prefix it must reproduce the batch quadratic form, evidence, and posterior to numerical tolerance. This is a regression test, not evidence for the joint model below.

F. PROPOSED JOINT HYBRID-DAE BAYES $\mathcal{E}_{\mathrm{J}}$

The proposed extension would infer state trajectory X, algebraic variables Z, event family c, onset τ, support S, amplitudes a_S, operating nuisance $\boldsymbol{\eta}$, and integrity corruption b. Its posterior factorization is

$$\begin{aligned} p(X, Z, \boldsymbol{\eta}, b, c, \tau, S, a_S \mid Y) &\propto p(\boldsymbol{\eta}) p(c) p(\tau \mid c) p(S \mid c) \\ &\times p(a_S \mid S, c) p(\boldsymbol{\chi}_0 \mid \boldsymbol{\eta}) \\ &\times \prod_{t=0}^{T-1} p(\boldsymbol{\chi}_{t+1} \mid \boldsymbol{\chi}_t, \boldsymbol{\eta}, c, \tau, S, a_S) \\ &\times \prod_{t=0}^{T} p(y_t \mid \boldsymbol{\chi}_t, b_t, c, \tau, S, a_S) p(b_t), \end{aligned} \tag{148}$$

subject to the candidate DAE constraints. Here $\boldsymbol{\chi}_t$ denotes a consistent state/nuisance chart. A constrained negative twice-log density, excluding the discrete candidate priors, is

$$\begin{aligned} \mathcal{J}_S^\circ = \|\boldsymbol{\chi}_0 - \overline{\boldsymbol{\chi}}_0\|^2_{P_0^{-1}} + \sum_{t=0}^{T} \|r_{S,t}^{\mathrm{pmu}}\|^2_{R_t^{-1}} \\ + \sum_{t=0}^{T-1} \|r_{S,t}^{\mathrm{dyn}}\|^2_{Q_t^{-1}} + 2\lambda_b \sum_{t=0}^{T} \|b_t\|_1. \end{aligned} \tag{149}$$

Candidate evidence must be evaluated in a common feasible measure. The boundary correction is exposed by an explicit concentration parameter rather than by informally assuming a concentrated posterior.

Let $\mathcal{R}_0$ fix operating point, onset, and load family and replace the candidate prediction with the same quadratic manifold, W2 covariance, support prior, and amplitude prior used by $\mathcal{E}_{\mathrm{B}}$. Then the defining regression condition is

$$p_{\mathrm{J}}(S, a_S \mid Y, \mathcal{R}_0) = p_{\mathrm{B}}(S, a_S \mid r, \boldsymbol{\eta}_0, \tau_0, c_{\mathrm{load}}). \tag{150}$$

The equality is not claimed for an unrestricted nonlinear DAE likelihood; it holds only when the complete statistical contract is shared.

Theorem 18 (Local cone–Laplace evidence): Let $L_{S,\epsilon}$ be the log likelihood plus the log of a proper prior in identifiable

TABLE 10. Evaluated 137-hypothesis Bayesian load-support algorithm.

Step	Operation	Computation and frozen choice	Status
0	Declare contract	Fix 16 candidate load buses, $K \leq 2$, known onset, 30-frame window, eight PMUs, nominal topology, and no corrupted channels	Evaluated
1	Validate plant	Check equilibrium, terminal-current orientation, sampling, nonlinear solve acceptance, and first-order response before building the bank	Evaluated
2	Build physical dictionary	Generate centered D_i, self-curvature Q_i, and cross-interaction Q_{ij} from simulator-consistent interventions	Evaluated
3	Fit noise on CAL	Estimate channel covariance and AR(1) dependence from normal calibration records; freeze W2 before event TEST	Evaluated
4	Freeze priors and integrator	Set $p(K)$, uniform support priors, Gaussian amplitude slab, GH31 nodes/weights, and independent GK checks	Evaluated
5	Form the residual	Align the known onset, subtract the frozen normal mean, retain the fixed PMU/channel order, and apply the declared temporal covariance	Evaluated
6	Enumerate supports	Construct H_0, 16 single-source supports, and 120 two-source supports; no learned source label prunes the bank	Evaluated
7	Integrate each support	Evaluate (136) with (137); retain quadrature-weighted amplitude moments	Evaluated
8	Normalize once	Add the support prior in log space, subtract the global log-sum-exp, and obtain (138)	Evaluated
9	Read decisions in order	Report event presence, $p(K \mid r)$, conditional support ranking, source inclusion, and conditional amplitude intervals separately	Evaluated
10	Quantify ambiguity	Compute entropy, effective multiplicity, or $\mathcal{C}_q$ without relabeling them as validated abstention guarantees	Defined; partial evidence
11	Record provenance	Store the plant identifier, support, amplitude, noise seed, likelihood version, and frozen artifact hash for every independent record	Evaluated

local coordinates of dimension d_S. Suppose the rescaled feasible domains converge locally to a closed polyhedral cone K_S and

$$L_{S,\epsilon}(\boldsymbol{\xi}_0 + \epsilon u) - L_{S,\epsilon}(\boldsymbol{\xi}_0) = h_S^{\mathsf{T}} u - \tfrac{1}{2} u^{\mathsf{T}} H_S u + r_\epsilon(u), \quad (151)$$

where $H_S \succ \mathbf{0}$ and $r_\epsilon \to 0$ uniformly on compact sets. Assume an integrable dominating envelope and negligible evidence from distant modes. Then

$$Z_{S,\epsilon} = e^{L_{S,\epsilon}(\boldsymbol{\xi}_0)} \epsilon^{d_S} (2\pi)^{d_S/2} |H_S|^{-1/2} \times \exp\left(\tfrac{1}{2} h_S^{\mathsf{T}} H_S^{-1} h_S\right) \Pr(U_S \in K_S)[1 + o(1)], \quad (152)$$

$$U_S \sim \mathcal{N}(H_S^{-1} h_S, H_S^{-1}).$$

a: Proof.

Set $\boldsymbol{\xi} = \boldsymbol{\xi}_0 + \epsilon u$, which contributes the Jacobian ϵ^{d_S}. The compact-uniform expansion gives pointwise convergence of the rescaled integrand to $\exp(h_S^{\mathsf{T}} u - u^{\mathsf{T}} H_S u/2)$. The assumed envelope permits dominated convergence, while the distant-mode condition localizes the original integral. Completing the square in the limiting quadratic integral gives the Gaussian normalizer and the cone probability in (152). □

For an interior model $K_S = \mathbb{R}^{d_S}$ and the probability factor is one. For one or more boundary amplitudes it is a Gaussian orthant probability, not generally 2^{-q}. Half-normal prior normalizers supply their own factors in addition to this cone probability. Chart Jacobians, prior normalizers, and candidate-dependent covariance determinants belong in $L_{S,\epsilon}$ or the measure term. Comparing only optimized residuals drops both volume and boundary terms and does not produce a posterior.

For a small-noise Gaussian Hybrid-DAE trajectory map $\boldsymbol{\mu}_S(\boldsymbol{\xi}_0 + \epsilon u) = \boldsymbol{\mu}_0 + \epsilon J_S u + O(\epsilon^2)$,

$$h_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} z, \qquad H_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} J_S, \quad (153)$$

so positive curvature is equivalent to first-order identifiability in the reduced coordinates. This is a theoretical local specialization, not a validation of the unevaluated joint estimator. If H_S is singular, merely replacing its inverse by a pseudoinverse is invalid: redundant coordinates must first be quotiented out, while genuine higher-order identification changes the evidence scale. For example,

$$\int_0^\infty e^{-na^4/4}\, da = \frac{\Gamma(1/4)}{4^{3/4}} n^{-1/4}, \quad (154)$$

not the $n^{-1/2}$ rate of regular quadratic Laplace theory.

Proposition 3 (Exact reduction along redundant fibers): Suppose a parameter chart $\boldsymbol{\theta} \leftrightarrow (\boldsymbol{\xi}, \boldsymbol{\zeta})$ is one-to-one on the prior support and the observable map satisfies $\boldsymbol{\mu}(\boldsymbol{\theta}) = \overline{\boldsymbol{\mu}}(\boldsymbol{\xi})$ there. If $\widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})$ is the transformed proper prior, then the evidence equals the reduced integral with induced prior

$$\overline{\pi}(\boldsymbol{\xi}) = \int \widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})\, d\boldsymbol{\zeta}. \quad (155)$$

Thus an exactly redundant coordinate contributes no vanishing small-noise power; its effect is the pushforward prior measure, not a pseudodeterminant.

b: Proof.

The likelihood is independent of $\boldsymbol{\zeta}$. Fubini's theorem integrates the proper transformed prior along each fiber and gives $Z = \int p(y \mid \boldsymbol{\xi}) \overline{\pi}(\boldsymbol{\xi}) d\boldsymbol{\xi}$ exactly. □

For a merely local chart, the same reduction is asymptotic only after a localization condition excludes evidence from distant branches.

Theorem 19 (Mixed regular and higher-order evidence): In whitened coordinates, suppose the first observable expansion after redundancy removal is

$$\boldsymbol{\mu}(\boldsymbol{\xi}, a) = \boldsymbol{\mu}_0 + A\boldsymbol{\xi} + \frac{c_k}{k!} a^k + o(\|\boldsymbol{\xi}\| + |a|^k), \quad (156)$$

where A has rank r, the proper prior density is continuous and positive at the origin, and

$$c_{k,\perp} = (I - P_A) c_k \neq \mathbf{0}. \quad (157)$$

TABLE 11. Deterministic checks of boundary-aware evidence identities. These toy systems are not evaluated estimator results.

Check	Recomputed outcome
Signed/one-sided Bayes factors	max quadrature error 2.7e-12 (pass).
Broad nuisance limit	relative information error 2.1e-09 (pass).
Boundary likelihood ratio	zero mass 0.4972; positive mean 0.9987 (pass).
Nonlinear cone–Laplace	relative-error slope 0.9497 (pass).
Singular quartic evidence	scaling slope -0.2500 (pass).

Assume a dominating envelope and negligible evidence away from this branch. For the Gaussian-normalizer-free evidence kernel with noise scale ϵ,

$$M_\epsilon \sim C\epsilon^{r+1/k}, \tag{158}$$

$$C = \pi(\mathbf{0},0)(2\pi)^{r/2}|A^{\mathsf{T}}A|^{-1/2} \tag{159}$$

$$\times \frac{\Gamma(1/(2k))}{k}\left[\frac{2(k!)^2}{\|c_{k,\perp}\|_2^2}\right]^{1/(2k)}. \tag{160}$$

c: Proof.

Use $\boldsymbol{\xi} = \epsilon u$ and $a = \epsilon^{1/k}t$. The Jacobian is $\epsilon^{r+1/k}$ and the limiting kernel is $\exp[-\|Au + c_k t^k/k!\|^2/2]$. The component of c_k in $\operatorname{range} A$ disappears after a translation of u. Gaussian integration over the remaining regular coordinates yields $(2\pi)^{r/2}|A^{\mathsf{T}}A|^{-1/2}$; integration of the normal component over $t \in \mathbb{R}$ gives the gamma-function factor. Dominated convergence supplies the prior density at the origin and the stated limit. □

For one additional amplitude, the theorem separates three cardinality cases. A regular direction supplies an $O(\epsilon)$ Occam factor, a direction first visible at order k supplies $O(\epsilon^{1/k})$, and an exactly redundant extension supplies an $O(1)$ factor determined by its induced prior. For $\epsilon \asymp n^{-1/2}$ these become $n^{-1/2}$, $n^{-1/(2k)}$, and n^0. These are local evidence rates; the RAWSIM39 gate ledger does not contain the higher-order derivatives required to assign them to its rejected parameters.

For a sliding window, eliminating old Gaussian variables creates an arrival prior. Write its canonical log kernel as $-\frac{1}{2}z^{\mathsf{T}}\Lambda z + h^{\mathsf{T}}z + c$. If the information matrix and vector are partitioned into variables a to eliminate and retained variables b, exact integration yields

$$\Lambda_{\text{new}} = \Lambda_{bb} - \Lambda_{ba}\Lambda_{aa}^{-1}\Lambda_{ab}, \tag{161}$$

$$h_{\text{new}} = h_b - \Lambda_{ba}\Lambda_{aa}^{-1}h_a, \tag{162}$$

$$c_{\text{new}} = c + \tfrac{1}{2}h_a^{\mathsf{T}}\Lambda_{aa}^{-1}h_a + \tfrac{n_a}{2}\log(2\pi) - \tfrac{1}{2}\log\det\Lambda_{aa}. \tag{163}$$

The first two lines preserve the next MAP problem; the last line is also required to preserve evidence between candidates.

Table 12 gives the intended sequence. It is a specification for the next experiment, not a claim about the results reported in the main paper.

For a posterior state $\boldsymbol{\pi}_t(S) = p(S \mid Y_{1:t})$, an idealized unit-cost support decision with per-frame delay cost c satisfies the Bellman equation

$$V(\boldsymbol{\pi}) = \min\left\{1 - \max_S \pi(S),\ c + \mathbb{E}[V(\boldsymbol{\pi}_{t+1}) \mid \boldsymbol{\pi}_t = \boldsymbol{\pi}]\right\}. \tag{164}$$

The one-step expected information gain is

$$I(S; y_{t+1} \mid Y_{1:t}) = H(S \mid Y_{1:t}) - \mathbb{E}[H(S \mid Y_{1:t+1})]. \tag{165}$$

Equations (164)–(165) motivate adaptive horizon selection. They are not an executed stopping rule for the current fixed-onset, fixed-window bank.

Finally, a hidden-voltage or derived-output posterior must average across support uncertainty. With $w_S = p(S \mid Y)$, conditional output mean m_S, and covariance P_S,

$$\overline{m} = \sum_S w_S m_S, \tag{166}$$

$$\operatorname{Cov}(\psi \mid Y) = \sum_S w_S P_S + \sum_S w_S (m_S - \overline{m})(m_S - \overline{m})^{\mathsf{T}}. \tag{167}$$

The second term is between-support ambiguity. Nonlinear output moments may be computed from posterior samples or a local delta method; phasors are averaged in rectangular coordinates before magnitude and angle are formed.

G. ESTIMATOR CLAIM BOUNDARY

The state recursion is an independently evaluated component. $\mathcal{E}_{\mathrm{B}}$ computes normalized Bayesian evidence under the frozen quadratic W2/L2 model and is the implemented reference event estimator. $\mathcal{E}_{\mathrm{J}}$, its reduction test, learned discrepancy, unknown-onset/family inference, changing-PMU operation, virtual outputs, and sequential stopping remain specified but unexecuted. The physical recentering is a regularized constrained estimator with separately assessed uncertainty; it is not relabeled as either event estimator.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5}. Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

TABLE 12. Candidate-conditioned joint estimator design and current implementation boundary.

Step	Operation	Required calculation	Present status
1	Pre-event update	Use only pre-event measurements to update state and operating nuisance; retain cross covariance and normalized arrival evidence	State filter evaluated; joint nuisance pending
2	Integrity mask	Mark missing channels exactly and introduce sparse corruption variables for available but suspect channels	Ontology defined; common inference pending
3	Candidate generation	Enumerate physical family, support, onset, and severity domain without using source identity as a learned label	Load family evaluated; other families pending
4	Candidate prediction	Apply the candidate DAE/event map and the current PMU membership to generate a physical trajectory	Load bank evaluated
5	Candidate-conditioned solve	Fit state and nuisance on the feasible DAE manifold, conditional on the physical candidate and amplitude quadrature node	Not executed jointly
6	Learned discrepancy	If used, predict only candidate-residual discrepancy and covariance; do not replace the physical hypothesis or consume target source labels	Not implemented
7	Evidence normalization	Retain GH31 amplitude marginalization when $\lvert S\rvert \leq 2$; integrate or approximate the remaining coordinates with all determinants, priors, and Schur normalizers	GH31 load evidence evaluated; joint evidence pending
8	Posterior outputs	Return state-functional mixture, cardinality, support, integrity state, severity, and candidate-set uncertainty	Separate state/load outputs only
9	Sequential decision	Continue observing when expected information gain justifies delay; use distinct thresholds for detection, cardinality, and exact source	Not evaluated
10	Reconfigure and verify	After accepting a physical change, rebuild the operating center and check subsequent innovations before declaring a persistent model update	Static recentering only

TABLE 13. Complete frozen-contract inventory. The main paper groups these contracts by research question.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh H_0 + 16 singles +120 pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument. The two compact tables first preserve the exact mismatch and weak-signal values; the full-width panels then show the corresponding temporal, likelihood, and supervised-baseline diagnostics.

TABLE 14. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

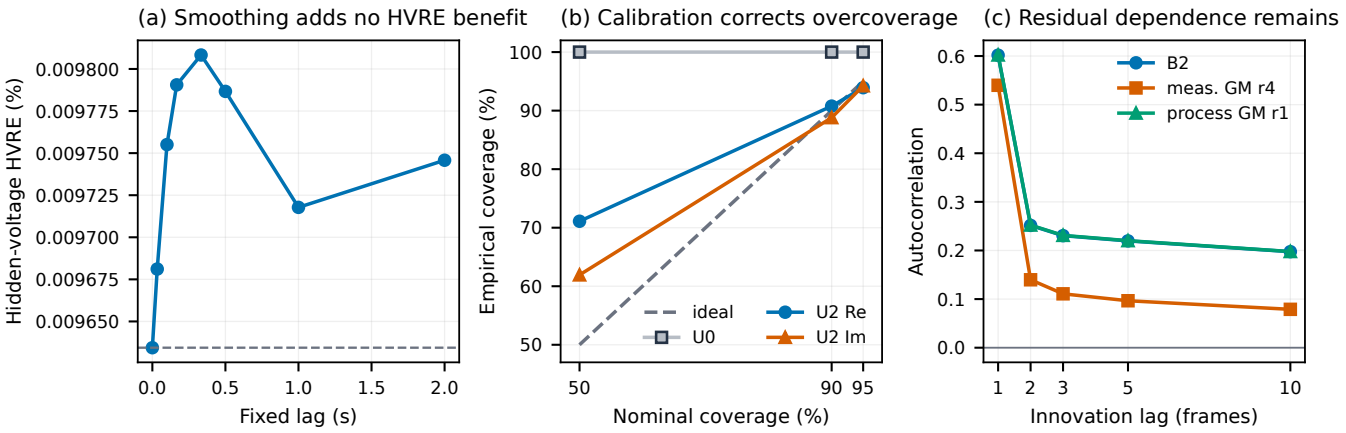


FIGURE 1. **Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.**

TABLE 15. **Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.**

$\lvert a \rvert$ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

TABLE 16. **Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.**

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

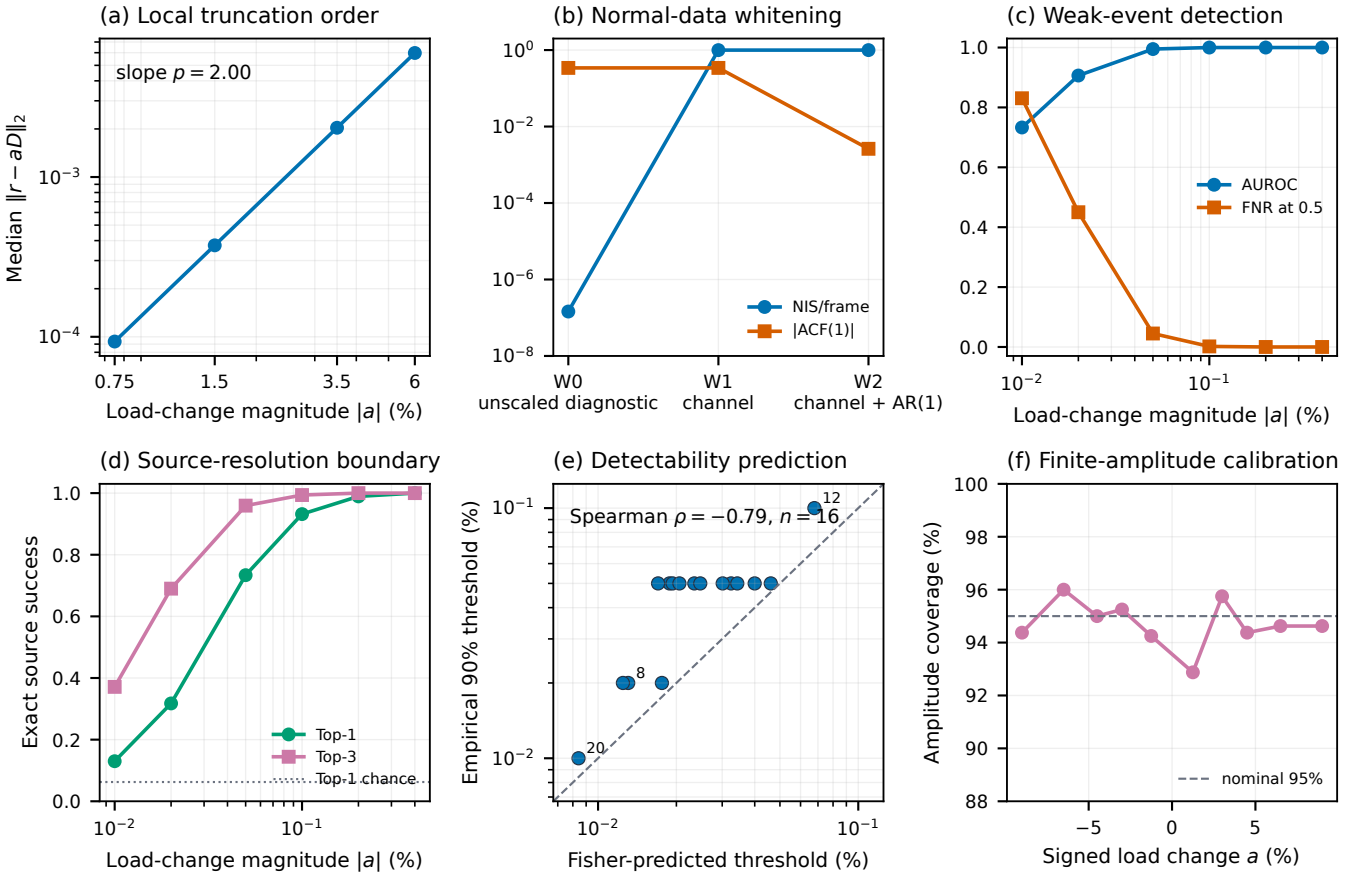


FIGURE 2. **Single-source development evidence for the frozen load likelihood. The figure is retained as component validation; GLOBAL-137 in the main paper is the prospective support-recovery result.**

TABLE 17. **Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.**

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

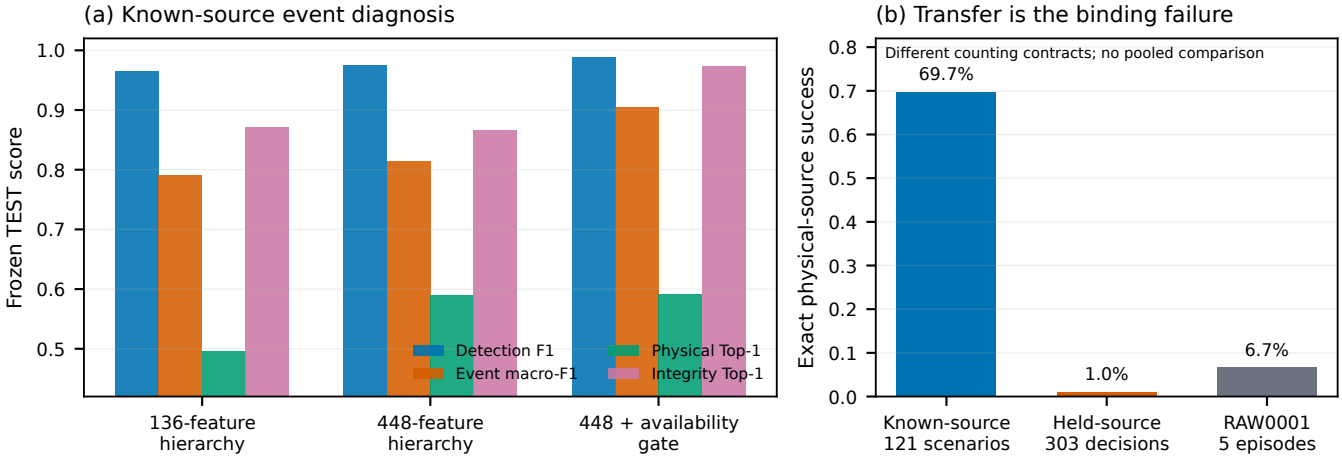


FIGURE 3. **Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.**

APPENDIX H EXPLORATORY HIERARCHICAL SINGLE-EVENT PILOT

This appendix records the September 24, 2026 estimator handoff as a *provisional* numerical branch of the joint design $\mathcal{E}_{\mathrm{J}}$. Its source documents are the 30-page mathematical derivation book and 24-page technical integration dossier identified in the repository integration ledger. Their statements are technical source material, not independent instructions or substitutes for campaign outputs. The dossiers label many checks "artifact-verified" in packages supplied elsewhere; those packages were not supplied with this manuscript checkout. The latest strict-final comparison is explicitly tagged terminal-reported in both documents. All values below are therefore transcribed as *dossier-reported* diagnostics and excluded from the article's frozen primary tables.

A. STATE, MEASUREMENTS, AND EVENT OPERATORS

The pilot's declared model has 114 differential states, 78 algebraic rectangular-voltage coordinates, and 32 measured rectangular channels: voltage at eight PMUs and terminal current at eight monitored branches. Sampling is at $1/30$ s. Its default matrices are $P_0 = 10^{-2} I_{114}$, $Q = 10^{-6} I_{114}$, and $R = 10^{-6} I_{32}$. These numbers identify the pilot likelihood, not the separately frozen covariance W2 of $\mathcal{E}_{\mathrm{B}}$. Three-phase phasors, frequency, ROCOF, and data-present fields in the historical competition files do not silently become inputs to this 32-channel physical likelihood.

For a fixed candidate H and severity ϑ_H, the semi-explicit system is $\dot{x} = f_H(x, z; \vartheta_H)$, $\mathbf{0} = g_H(x, z; \vartheta_H)$, and $y = h_H(x, z) + v$. At a regular point with invertible $G_z = \partial g_H / \partial z$, the implicit-function theorem gives

$$Z_x = -G_z^{-1} G_x, \quad Z_\vartheta = -G_z^{-1} G_\vartheta, \quad A_c = F_x - F_z G_z^{-1} G_x. \tag{168}$$

The corresponding reduced event pathway is $B_\vartheta = F_\vartheta + F_z Z_\vartheta$ and the direct measured pathway is $D_\vartheta = H_\vartheta + H_z Z_\vartheta$. A generator-reference change can satisfy $D_\vartheta = \mathbf{0}$ while $B_\vartheta \neq \mathbf{0}$: it is invisible to a raw algebraic screen but visible after dynamic propagation. The event operator and its duration must be replayed consistently in truth generation and candidate fitting. The pilot dossier reports a five-frame fault pulse, persistent load steps unless recovery is declared, and finite generator and line events with recovery. Some earlier inference runs used persistent approximations for the latter two; those rankings have a known model-contract discrepancy.

The current single-event catalog has 39 fault buses, 19 load buses, nine controlled generators, 46 in-service branches, and a separate H_0 comparison. Thus it has 113 physical candidates and one null candidate; the screening rank denominators may be smaller when a candidate is numerically unscorable. Fault severity uses a log-resistance coordinate, load severity two active/reactive changes, generator severity a reference-power change, and an outage has a discrete branch and switching profile. Missing or bad PMU channels belong to the integrity variable of $\mathcal{E}_{\mathrm{J}}$ and were not jointly inferred in this pilot.

B. JOINT GAUSS–NEWTON STEP AND ITS INTERPRETATION

For one candidate, write the conditional MAP residual as $r_H(X, \vartheta)$ after whitening and appending proper prior residuals. Let $\Delta u = \mathrm{col}(\Delta X, \Delta\vartheta)$ and $g = \mathrm{col}(g_X, g_\vartheta)$. At an iterate, its local quadratic is

$$q(\Delta u) = g^{\mathsf{T}} \Delta u + \tfrac{1}{2} \Delta u^{\mathsf{T}} H \Delta u, \qquad H = \begin{bmatrix} H_{XX} & H_{X\vartheta} \\ H_{\vartheta X} & H_{\vartheta\vartheta} \end{bmatrix}. \tag{169}$$

Eliminating ΔX gives $q_{\mathrm{red}}(\Delta\vartheta) = \mathrm{const} + b_H^{\mathsf{T}} \Delta\vartheta + \tfrac{1}{2} \Delta\vartheta^{\mathsf{T}} S_H \Delta\vartheta$, where $S_H = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ and $b_H = g_\vartheta - H_{\vartheta X} H_{XX}^{-1} g_X$. Hence its unconstrained local decrease is $-\tfrac{1}{2} b_H^{\mathsf{T}} S_H^{-1} b_H$ when $S_H \succ 0$. This algebra needs no conditional convergence of X: it is the full-block Gauss–Newton step expressed with a smaller parameter solve. By contrast, if $\psi(\vartheta) = \min_X J_H(X, \vartheta)$, the identity $\nabla^2 \psi = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ is a local *profile-Hessian* statement only at a regular conditional optimum, and for the true Hessian rather than an arbitrary Gauss–Newton approximation. An unsettled state trajectory can change the sign of the reduced gradient without invalidating Schur elimination.

The numerical realization factors the temporally sparse state block and obtains its action on g_X and the few columns of $H_{X\vartheta}$ by solves; it does not form H_{XX}^{-1}. One Armijo step length updates both state and severity and evaluates the actual nonlinear DAE objective. The pilot selected Rodas4P for transition evaluation after the reported cross-integrator audit. That choice is a numerical guard, not a new inference theorem. Levenberg–Marquardt remains an independent finalizer candidate for the unresolved fault/line fits [23], [24].

C. DETECTION, PHYSICAL SCREENING, AND RACING

Let $\boldsymbol{\nu}_k = y_k - \hat{y}_{k|k-1}^{H_0}$ be the observed innovation. An alarm tests the adequacy of H_0 on measured channels; it is not itself the physical onset. The pilot refines onset over a bounded interval and fits the common initial state $\widehat{X}_{\mathrm{pre}}$ on data before that onset. Every candidate starts from this same reference. This prevents a strong event from contaminating its own null-model baseline.

For a smooth candidate nested at H_0, the efficient local score is $T_H = b_H^{\mathsf{T}} S_H^{-1} b_H$ under positive definite S_H. If a severity is physically one-sided, maximizing the local quadratic over its tangent cone replaces the unconstrained inverse step and changes the null law; the ordinary signed score is then only a proposal proxy. For a fault or topology switch, let s_H be its short-horizon physical response from $\widehat{X}_{\mathrm{pre}}$ and r the corresponding null residual. With $W^{\mathsf{T}} W = R_H^{-1}$ and nuisance projector P_N, define $\widetilde{s}_H = (I - P_N) W s_H$ and $\widetilde{r} = (I - P_N) W r$. A signed matched score is $(\widetilde{r}^{\mathsf{T}} \widetilde{s}_H)^2 / \|\widetilde{s}_H\|^2$, with the allowed severity domain imposed before interpreting it as a physical fit. The two screening scores need not share a calibrated scale across families.

The frozen pilot consequently ranks within each family, retains a family quota, and allocates increasing nonlinear

TABLE 18. Single-event pilot ledger transcribed from the two author-supplied September 24 dossiers. All rows require the corresponding campaign archives for an independent paper audit.

Check or campaign	Dossier-reported result and precise interpretation	Source tier
Linear Gaussian oracle E0	RTS and batch MAP means agree near 10^{-16}; covariance/log-likelihood checks near 10^{-14} after an omitted evidence constant was repaired. This guards the numerical algebra only.	Dossier AV
Three-node and five-bus toys	E1: five development and 30 evaluation seeds, hidden algebraic two-sigma coverage 0.927 versus nominal 0.954. E2: first-order classification 0.2875 over 10 hypotheses × eight repetitions; not a nonlinear full-catalog result.	Dossier AV
Transition/derivative audit	Rodas4P/Rodas5P: 23 shared successful candidates with maximum endpoint difference 4.9×10^{-7}; Rodas4P evaluated three of three selected Rodas5P failures. Finite-difference coverage was 25/25; one random direction retained a moderate derivative discrepancy.	Dossier AV
Joint update and severity	Full-block and Schur steps agree in 30/30 synthetic block systems to numerical precision. Four fault-resistance initializations of 0.01, 0.03, 0.05, and 0.08 ended in 0.02006–0.02133 around truth 0.021; only the first two passed the stated strict stationarity gate.	Dossier AV
Small hidden-event bank	A hidden LOAD@16 and LINE(23,24) ranked first against selected decoys after joint severity and line-index corrections. These were not full-catalog tests.	Dossier AV
H_0 and onset pilots	Fifty-one disjoint normal calibration windows yielded no alarms at two reported thresholds on that same set; this is neither independent alarm testing nor false alarms per hour. Blind onset alternatives differed by 26–27 frames in two cases.	Dossier AV
Stage-A development	Four cases; 451/456 attempted physical scores were valid (98.9%); family-stratified shortlist retained 4/4 truths.	Dossier AV
Blind mini-holdout	FAULT@20, LOAD@28, GEN@38, and LINE(4,14): within-family truth ranks 3/38, 1/19, 1/9, and 1/46. The shortlist and subsequent racing retained 4/4 truths; GEN survived at the six-candidate cutoff. Four quiet windows were a proposal-level control only.	Dossier AV
Strict final comparison	The later terminal summary reports 0/4 exact winners after 24 final fits. Fault and line include unconverged truth/competitors; load and generator have better-scoring converged neighbors. This is a failure audit, not an identifiable physical error rate.	TR; final archive absent

budgets $B1, B2, B3, B5, B10$ before strict finalization. A candidate with a failed score is recorded as unscorable rather than inserted into a tie by catalog order. The dossier reports that ranking was nonmonotone with iteration budget and that the generator truth survived the blind racing cutoff at rank six of six. Thus survival is a search diagnostic; it says nothing by itself about a final converged winner or computational savings. In the reported racing run all 21 candidates were still evaluated to ten iterations, so a wall-clock advantage has not been demonstrated.

The operational order of this *pilot* is fixed: (1) process only the declared 32-channel PMU sequence and evaluate the H_0 innovation; (2) record alarm time, refine physical onset candidates, and fit a common pre-event state; (3) score every admissible physical family/location with its own event operator and explicit numerical status; (4) retain the frozen within-family quotas; (5) apply the same staged nonlinear budgets to all survivors while checkpointing objective and stationarity; (6) finalize every remaining candidate to the same strict gate; and (7) compare only validly converged modes with consistent priors, normalizers, and boundary treatment. Outputs before step (6) are candidate sets and diagnostics, not final source decisions. This route is retrospective whenever onset refinement or smoothing uses samples after the decision time; it has no measured end-to-end latency guarantee.

Several discarded results in the dossier were caused by implementation changes rather than by power-system information limits. A generator screen based only on the algebraic Jacobian missed its differential pathway; a previous joint update computed but did not apply the severity step; and an outage candidate was once switched at frame zero instead of the inferred onset. A full-window H_0 fit absorbed an event into the null reference, while independent caps on state and severity steps invalidated a quadratic proposal score. Unscored fault candidates inherited catalog order through a tied sentinel value. One detection-delay calculation used the window start rather than its end, and some failed integrator returns were encoded as nonfinite values despite finite unfinished states. The corrected pilot treats each as a separate numerical or semantic gate. Its four-case truth survival must not be merged with superseded rankings from those earlier variants.

D. EVIDENCE AND THE FINAL COMPARISON GATE

For converged, regular, identifiable interior modes only, the local Laplace approximation is

$$\log p(Y \mid H) \simeq -J_H(\widehat{u}_H) + \tfrac{d_H}{2}\log(2\pi) - \tfrac{1}{2}\log|\nabla^2 J_H(\widehat{u}_H)|, \tag{170}$$

where u_H uses a common feasible measure and J_H includes all model-dependent prior and likelihood normalizers. The normalizer, prior scale, curvature determinant, and any boundary or quotient correction must be retained when comparing families of different dimension. Section E-F derives the boundary-aware alternative. A candidate that stops at $B10$ or violates the strict stationarity/solver gate cannot be made comparable merely by evaluating (170). The pilot has not demonstrated exact normalization against the load-bank restriction $\mathcal{R}_0$.

E. DOSSIER-REPORTED NUMERICAL LEDGER

Table 18 preserves the numerical chronology without treating small pilot denominators as population rates. "Dossier AV" means the PDF claims an external campaign package supports the entry; it does not mean this repository has reopened that package. "TR" is the dossier's own terminal-report tier.

The case-specific terminal report places converged LOAD@27 and LOAD@26 ahead of true LOAD@28. It quotes approximate objectives -29986.9, -29985.9, and -29980.7, respectively, with evidence values around 4609.98, 4610.22, and 4607.17. Those values are deliberately absent from the main manuscript: their sign convention and normalizing terms must be checked against the final archive before quantitative model comparison. No exact generator

margin was supplied. The fault and line comparisons are not physically interpretable until both truth and nearest competitors pass a common strict convergence gate.

F. REQUIRED CLOSURE BEFORE A MULTI-FAMILY CLAIM

The next frozen audit must first replay each event operator, measurement conversion, onset, duration, and initial state under the same truth and inference model. It must then use an independent finalizer for the fault/line candidates, retaining every failed solve in the denominator. For converged load/generator mislocalizations, successive oracles should fix the true pre-event state, timing, severity, and trajectory, then remove noise and add a counterfactual nearby PMU. Only a persistent small whitened truth-to-competitor margin after those controls supports an information-limited explanation. The final-survivor archive must supply per-candidate status, objective, prior and likelihood constants, Hessian measure, and evidence normalization. A larger source-disjoint holdout, operational normal exposure, and the $\mathcal{R}_0$ reduction test are still needed. The current pilot cannot be used to upgrade the separate $\mathcal{E}_{\mathrm{B}}$ load-support result into a validated multi-family joint estimator.

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